

A Riemannian Factor Model for Manifold-Valued Time Series*

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Abstract

We propose a Riemannian factor model (RFM), a novel framework for analyzing potentially high-dimensional time series data observed on Riemannian manifolds. Such time series are encountered in various applications, including economics, finance, medical imaging, and genomics and microbiome research. The proposed model is geometry-aware and accounts for the inherent nonlinearity in the data. In a high-dimensional asymptotic regime, where the manifold dimension is allowed to diverge with the sample size n , we establish convergence rates for the estimated loading space. In particular, under short-memory and strong factor conditions, we obtain a dimension-free $n^{-1/2}$ rate, which matches the convergence rate of the high-dimensional linear factor model. Finite-sample performance of the proposed RFM is demonstrated with simulated time series on the Bures–Wasserstein manifolds and products of spheres, as well as an application to monthly realized covariances of selected U.S. stock returns—modeled as time series in the Bures–Wasserstein manifold, where the RFM provides demonstrably interpretable factors and yields competitive predictive performance.

Keywords: Bures–Wasserstein metric, compositional time series, covariance prediction, dimension reduction, non-Euclidean time series

*Rong Chen’s research was supported in part by National Science Foundation grants DMS-2319260 and DMS-2413858. Yaqing Chen’s research was supported in part by National Science Foundation grant DMS-2311035.

1 Introduction

In recent years, time series that intrinsically live on a nonlinear manifold have emerged as an important type of data in many scientific fields. For example, compositional time series can be treated as sphere-valued data after the square root transformation (Wang et al. 2007, Dai & Müller 2018, Zhu & Müller 2024). As another example, distribution-valued data, such as the distributions of age-at-death, housing price (Chen et al. 2023), stock returns (Zhang et al. 2022) and the spatial distribution of taxi demand (Dai 2022), are also naturally nonlinear. Similarly, observations of symmetric positive-definite (SPD) matrices, common in medical imaging (e.g., diffusion tensor imaging), can be analyzed as data on some nonlinear manifold (Moakher 2005, Pennec et al. 2006, Arsigny et al. 2007, Harandi et al. 2018). SPD-valued data are also ubiquitous in finance, where time-varying covariance matrices play a key role (Johansson et al. 2023).

In addition to the intrinsic nonlinearity, the analysis of manifold-valued time series is often faced with high dimensionality. Indeed, dimension reduction of non-Euclidean data has been a topic of great interest and was studied under a few different settings. For instance, Fletcher et al. (2004) proposed the principal geodesic analysis (PGA), which performs linear PCA on the tangent space of the Fréchet mean—a generalization of the mean to metric space-valued random elements (Fréchet 1948). Huckemann & Ziezold (2006) considered alternative principal geodesics that typically do not pass the Fréchet mean. For specialized nonlinear spaces, Jung et al. (2012) proposed the principal nested spheres (PNS) for data on hyperspheres and Huckemann et al. (2010) considered quotient spaces from isometric Lie group actions. See also Huckemann & Eltzner (2021) for a discussion on the difficulty of extending PNS to other manifolds. Dai & Müller (2018) and Lin & Yao (2019) considered dimension reduction for functional observations on a Riemannian manifold. Nevertheless, the aforementioned works either focus on independent samples of functions or lack statistical

guarantees. As a result, the rigorous development of a dimension reduction framework for Riemannian time series remains an important gap in the current literature.

Factor models (Bai & Ng 2002, 2008, Stock & Watson 2006, 2011, Lam et al. 2011, Lam & Yao 2012) are among the most widely used methods for studying the temporal dynamics of high-dimensional time series. However, manifold-valued data are often unamenable to existing tools that are predominantly designed for data in linear spaces. One fundamental obstacle is that vector addition and scalar multiplication are rarely well-defined in typical nonlinear spaces. Moreover, because the classical factor models extract latent factors by projection onto lower-dimensional linear subspaces, they are inherently misspecified and unsuitable for analyzing intrinsically nonlinear data.

In this work, we propose a new factor model for time series lying on a Riemannian manifold, referred to as the Riemannian factor model (RFM). The RFM postulates that the latent factors govern the variations of the data around the Fréchet mean. To estimate the RFM, we extract latent factors on the tangent space at the empirical Fréchet mean. Since the population and empirical Fréchet means differ in general, the true and estimated loadings lie in different tangent spaces. Therefore, the asymptotic analysis of the RFM is more involved than classical Euclidean factor models. While retaining the computational simplicity of applying classical factor models to tangent spaces, the proposed RFM provides geometry-aware prediction and enhanced interpretability by leveraging its intrinsic formulation. For predictions, the outputs of RFM are guaranteed to lie on the manifold whereas the predictions of the linear factor models do not necessarily respect the manifold constraint. For interpretation, one can examine the effect of an extracted factor by studying the geodesic trajectory on the manifold traveling in the direction of the corresponding loading vector, which may offer additional insight. In addition, we also employ a ratio estimator for selecting the number of factors. Section 2 formally introduces the Riemannian factor model, discusses

the related assumptions, and details the estimation procedure.

As a leading example, we model the realized covariances between selected U.S. stocks—viewed as data on the Bures–Wasserstein manifold. Understanding the dynamics of the covariances between stocks is central in a number of financial applications, including portfolio selection (Markowitz 1952, Yen & Yen 2014) and risk management (Tsay 2010). Existing techniques (Johansson et al. 2023) such as the exponential weighted moving average (EWMA), while offering reasonable estimates, provide little interpretation in what drives the changes in realized volatility. In contrast, our method, through factor modeling on the Bures–Wasserstein manifold, yields interpretable factors as well as competitive predictions. By examining the geodesics associated with the corresponding loading vectors, we found that the factors estimated by RFM capture market-wide and sector-specific effects, where the market factor closely traces the closing price of the VIX volatility index. Section 5 contains more detail on the real data analysis.

We establish consistency and convergence rates for the proposed RFM estimators in a high-dimensional regime where the manifold dimension is allowed to diverge with the sample size. In this setting, the theoretical challenges are twofold. First, the high-dimensional regime requires a refined analysis with non-asymptotic bounds on the estimation errors, including the statistical errors and the distortion caused by the mismatch between tangent spaces of the population and the empirical Fréchet means. Second, the analysis should account for the temporal dependence in the data, which is further complicated by the curved geometry of the underlying manifold. This is in sharp contrast to the existing literature on dimension reduction (Dai & Müller 2018, Lin & Yao 2019) that typically assumes the manifold dimension is fixed and data consist of independent manifold-valued measurements or functions. Specifically, the Taylor approximation approach employed in the prior works has an unfavorable dimension dependence, and, unlike the case for functional data where

independent copies of continuous sample paths are available, the temporal correlation among Riemannian time series has a direct impact on the rate of convergence. In addition, due to the tangent space mismatch, the estimated factor loading space cannot be directly compared with the true loading space without parallel transportation. To address these issues, we employ a novel technique by studying the holonomy, which bounds the change in the tangent vectors parallel-transported along a small geodesic triangle. In addition, under a short memory condition, a dimension-free $n^{-1/2}$ rate can still be obtained, where n denotes the sample size. A detailed discussion on the theoretical results is presented in Section 3.

The main contributions of this work are as follows. First, to the best of our knowledge, this work proposes not only the first factor model but also the first dimension reduction method with theory-backed estimation for Riemannian time series. In addition, unlike the PCA-type methods that hinge on the variance-covariance operator, our approach employs the autocovariance operator to estimate the factor loading space. Similar to the works of Peña & Box (1987), Peña & Poncela (2006), Lam et al. (2011), Lam & Yao (2012) for the Euclidean data, one key advantage of this approach is that it accommodates arbitrary cross-correlation in the noise, and avoids the identification issues in principal component-based factor models (Bai & Ng 2002, 2008, Bai & Li 2012, Forni et al. 2000, Stock & Watson 2006, 2011). Second, this paper is also among the first to apply factor modeling to data on the Bures–Wasserstein manifold. The Bures–Wasserstein metric, which is strongly motivated by optimal transport, has gained traction recently in various fields, including single-cell transcriptomics (Xu & Li 2025a,b, Haviv et al. 2025), neuroscience (Zirpoli et al. 2025), computational chemistry (Jiang et al. 2025), and diffusion tensor imaging (Tang et al. 2024). However, most prior works on dimension reduction for SPD-valued data have mostly focused on the affine-invariant (Moakher 2005, Pennec et al. 2006) and Log-Euclidean (Arsigny et al. 2007) geometries, except for the recent work by Santoro & Panaretos (2023) which studied

functional PCA for infinite-dimensional covariance operators. Third, rigorous theoretical guarantees for the estimation method and the selection of the number of factors are derived under the high-dimensional Riemannian time series setup, where novel proof techniques are needed.

The paper is organized as follows. In Section 2, the RFM is formally introduced, along with an efficient estimation procedure. Section 3 provides several theoretical properties of the proposed estimators. In Section 4, the performance of RFM is studied using synthetically generated time series on nonlinear manifolds, including the Bures–Wasserstein manifold and the product manifold of spheres. In Section 5, we apply the RFM to the monthly covariance matrices of selected U.S. stock returns, viewed as time series in the Bures–Wasserstein manifold. Section 6 concludes. All proofs to the theoretical results are presented in the supplementary material.

Next, we introduce some notations needed for the rest of the paper. As discussed above, we adopt the high-dimensional asymptotic regime in which the manifold dimension $p = p(n)$ may diverge with the sample size n , following the triangular array framework of [Bühlmann & van de Geer \(2011\)](#). Hence, for example, a function $f(p, n) = f(p(n), n)$ can be shorthand as $f(n)$. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be the probability space on which the triangular array is defined. For two sequences of random variables X_n, Y_n , we write $X_n = O_p(Y_n)$ if for any $\epsilon > 0$ there exists $C > 0$ and $N < \infty$ such that $\mathbb{P}(|X_n/Y_n| > C) \leq \epsilon$ for all $n \geq N$. Similarly, $X_n = o_p(Y_n)$ if for any $\epsilon > 0$, $\mathbb{P}(|X_n/Y_n| > \epsilon) \rightarrow 0$. If X_n, Y_n are deterministic, we use O and o in lieu of O_p and o_p , respectively. In addition, if $a_n = O(b_n)$ and if $b_n = O(a_n)$, we write $a_n \asymp b_n$.

2 The Riemannian factor model: Setup and estimation

In this section, we introduce the Riemannian factor model (RFM), which aims to capture the dynamics of manifold-valued time series through latent factors. We begin by reviewing some elements of Riemannian geometry. Then the RFM is introduced, along with discussions about some assumptions maintained throughout the paper. Finally, we present an estimation procedure for RFM.

2.1 Preliminaries

We first review some elements of Riemannian geometry and introduce notations. The reader can consult, for example, [do Carmo \(1992\)](#), for a more comprehensive treatment. See also [Bhattacharya & Patrangenaru \(2003, 2005\)](#) and [Lin & Yao \(2019\)](#) for brief introductions.

Let M be a connected p -dimensional Riemannian manifold with a Riemannian metric $\langle \cdot, \cdot \rangle_{\mathbf{x}}$, where $\mathbf{x} \in M$. For $\mathbf{x}, \mathbf{y} \in M$, the geodesic distance is defined as $d_M(\mathbf{x}, \mathbf{y}) := \inf_{\gamma} \int \langle \gamma'(t), \gamma'(t) \rangle_{\gamma(t)}^{1/2} dt$, where the infimum is over all piecewise smooth curves γ on M joining $\mathbf{x}$ and $\mathbf{y}$ and $\gamma'(t) = d\gamma/dt$ denotes the tangent vector field of γ . Let D be the (unique) Levi-Civita connection (see Theorem 3.6, Chapter 2, [do Carmo 1992](#)) associated with the Riemannian metric. A smooth curve $\gamma : [0, 1] \rightarrow M$ is called a geodesic if $D_{\gamma'(t)}\gamma'(t) = 0$, meaning that its acceleration vanishes with respect to the connection. Geodesics are locally length-minimizing curves and serve as the analogs of straight lines in the Euclidean space. The tangent space of M at a point $\mathbf{x}$ is denoted by $T_{\mathbf{x}}M$. Given a tangent vector $\mathbf{v} \in T_{\mathbf{x}}M$ with sufficiently small norm $\|\mathbf{v}\|_{\mathbf{x}} = \langle \mathbf{v}, \mathbf{v} \rangle_{\mathbf{x}}^{1/2}$, there exists a unique geodesic γ such that $\gamma(0) = \mathbf{x}$ and $\gamma'(0) = \mathbf{v}$. For such $\mathbf{v}$, the (Riemannian) exponential map at $\mathbf{x}$ is defined as $\text{Exp}_{\mathbf{x}}(\mathbf{v}) = \gamma(1)$. It is well known that there exists $R > 0$ such that $\text{Exp}_{\mathbf{x}}$ is diffeomorphic on $V_R = \{\mathbf{v} \in T_{\mathbf{x}}M : \|\mathbf{v}\| < R\}$, in which case the geodesic ball with radius R , $B_{\mathbf{x}}(R) = \{\mathbf{y} \in M : d_M(\mathbf{x}, \mathbf{y}) < R\}$, is called a normal neighborhood of $\mathbf{x}$. The

supremum of such R 's is called the radius of injectivity. Furthermore, if for every $\mathbf{y} \in B_{\mathbf{x}}(R)$ there exists a normal neighborhood $G_{\mathbf{y}}$ of $\mathbf{y}$ such that $G_{\mathbf{y}} \supset B_{\mathbf{x}}(R)$, then $B_{\mathbf{x}}(R)$ is called a totally normal neighborhood. It follows that every pair of points in the totally normal neighborhood can be connected by a unique length-minimizing geodesic. In the rest of the paper, when there is no danger of confusion, we may omit the subscript indicating the base point $\mathbf{x} \in M$ in the Riemannian metric and its norm, and write $\langle \cdot, \cdot \rangle_{\mathbf{x}} = \langle \cdot, \cdot \rangle$, $\| \cdot \|_{\mathbf{x}} = \| \cdot \|$.

Along a geodesic γ , a vector field $V(t) \in T_{\gamma(t)}M$ is said to be parallel if $D_{\gamma'(t)}V(t) = 0$ for all t . In fact, given $V_0 \in T_{\gamma(0)}M$, there exists a unique parallel vector field $V(t)$ along γ with $V(0) = V_0$. The tangent vector $V(t) \in T_{\gamma(t)}M$ is called the parallel-transported vector of V_0 from $\gamma(0)$ to $\gamma(t)$. If the geodesic is unique, then by uniqueness of the parallel vector field this defines the parallel transport operator $\mathcal{P}_{\mathbf{x} \rightarrow \mathbf{y}} : T_{\mathbf{x}}M \rightarrow T_{\mathbf{y}}M$.

The Riemannian curvature tensor $\mathcal{R}$ describes how the manifold bends. Formally, for vector fields X, Y, Z on M , $\mathcal{R}(X, Y)Z = D_X D_Y Z - D_Y D_X Z - D_{[X, Y]}Z$, where $[X, Y]$ is the Lie bracket of vector fields. Given linearly independent $\mathbf{u}, \mathbf{v} \in T_{\mathbf{x}}M$, the sectional curvature of the two-dimensional space $\sigma = \text{span}\{\mathbf{u}, \mathbf{v}\}$ is defined as

$$K(\mathbf{x}, \sigma) = \frac{\langle \mathcal{R}(\mathbf{u}, \mathbf{v})\mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2 - \langle \mathbf{u}, \mathbf{v} \rangle^2}.$$

The definition is independent of the choice of the basis $\{\mathbf{u}, \mathbf{v}\}$ for σ . Nonzero curvature fundamentally distinguishes a curved space from a flat one, such as the Euclidean space. Hence, it is crucial in both the study of Riemannian geometry and our theoretical analysis.

As a concrete example, the set of real symmetric positive-definite (SPD) matrices of the same dimensions, equipped with the Bures–Wasserstein metric, forms a Riemannian manifold (Malagò et al. 2018, Chewi et al. 2025, Takatsu 2011). The resulting geodesic distance between two SPD matrices Σ_1, Σ_2 is $\{\text{Tr}(\Sigma_1 + \Sigma_2 - 2(\Sigma_1^{1/2} \Sigma_2 \Sigma_1^{1/2})^{1/2})\}^{1/2}$. This corresponds to the 2-Wasserstein distance between the two normal distributions $N(0, \Sigma_1)$ and $N(0, \Sigma_2)$, which is the optimal transport cost for transferring mass from the distribution $N(0, \Sigma_1)$

to $N(0, \Sigma_2)$. Due to this connection to optimal transport, the Bures–Wasserstein metric appears to be more natural to econometric and financial applications compared to other Riemannian metrics, such as the affine-invariant (Moakher 2005, Pennec et al. 2006) or the log-Euclidean (Arsigny et al. 2007) metrics—in fact, optimal transport models are widely seen in economics and finance. For example, in portfolio selection, Blanchet et al. (2022) employ a distributionally robust approach where the ambiguity set is a Wasserstein ball (see also Pesenti & Jaimungal 2023). As another example, model-independent hedging is related to the martingale optimal transport problem (De Marco & Henry-Labordère 2015, Henry-Labordere 2017). Galichon (2016) provides an excellent overview of the applications of optimal transport in econometrics. In Section S.2 of the supplementary material, we briefly review the properties of the Bures–Wasserstein manifold. However, contrary to the affine-invariant and the log-Euclidean metrics, the Bures–Wasserstein manifold has nonnegative sectional curvature and is incomplete. Although these properties introduce some complexity to the theoretical analysis, our theoretical results are applicable in this setting; see Section 3 for further discussions.

2.2 The Riemannian factor model

We now introduce the Riemannian factor model (RFM) for time series taking values in a Riemannian manifold M . Let $\mathbf{x}_t \in M$ be the observed manifold-valued time series. The RFM assumes

$$\mathbf{x}_t = \text{Exp}_{\boldsymbol{\mu}}(\mathbf{z}_t), \tag{1}$$

where $\boldsymbol{\mu} \in M$ is a fixed reference point and the latent time series $\mathbf{z}_t$ is mean-zero and takes values in the p -dimensional tangent space $T_{\boldsymbol{\mu}}M$. Under some assumptions, $\boldsymbol{\mu}$ can be viewed as the Fréchet mean of $\mathbf{x}_t$ (See Remark 1 below). Intuitively, the latent time series $\mathbf{z}_t$ represents the deviation from $\boldsymbol{\mu}$, which is mapped to the observed $\mathbf{x}_t$ via the Riemannian

exponential. Then RFM specifies a factor model for $\mathbf{z}_t$ in the tangent space,

$$\mathbf{z}_t = \mathcal{A}(\mathbf{f}_t) + \boldsymbol{\delta}_t, \quad (2)$$

where $\mathbf{f}_t \in \mathbb{R}^r$ with $r \ll p$, is the factor process, $\boldsymbol{\delta}_t$ is white noise, and $\mathcal{A} : \mathbb{R}^r \rightarrow T_{\boldsymbol{\mu}}M$ is an isometric linear operator. Hence $\mathbf{z}_t$ is governed by only a few latent factors. Here we adopt the framework of Lam et al. (2011) and Lam & Yao (2012) where the factors account for the dynamic correlations, and it is assumed that no linear combination of $\mathbf{f}_t$ would result in a white noise. This allows the covariance of $\boldsymbol{\delta}_t$ to be arbitrary. Different from the PCA or PGA models adopted in the existing non-Euclidean statistics literature (Fletcher et al. 2004, Fletcher & Joshi 2007, Huckemann et al. 2010, Dai & Müller 2018, Lin & Yao 2019), our focus is on the dynamic correlations.

To illustrate the factor model formulation (2), fix an orthonormal basis $\mathbf{E} = (\mathbf{e}_1, \dots, \mathbf{e}_p)$ of $T_{\boldsymbol{\mu}}M$. We can express $\mathbf{z}_t$ as a vector $\mathbf{z}_{t,\mathbf{E}} \in \mathbb{R}^p$ with respect to $\mathbf{E}$. The linear factor model of Lam et al. (2011) assumes

$$\mathbf{z}_{t,\mathbf{E}} = \mathbf{A}_{\mathbf{E}}\mathbf{f}_t + \boldsymbol{\delta}_{t,\mathbf{E}}, \quad (3)$$

where $\mathbf{A}_{\mathbf{E}}$ is assumed to have orthonormal columns without loss of generality. The following proposition shows the two models (2) and (3) are equivalent.

Proposition 1. *If (2) holds with*

(F) $\{\boldsymbol{\delta}_t\}$ is a mean-zero white noise, and $\mathcal{A}$ is isometric,

then for any orthonormal basis $\mathbf{E}$ of $T_{\boldsymbol{\mu}}M$, $\mathbf{z}_{t,\mathbf{E}}$ satisfies (3) with

(F') $\boldsymbol{\delta}_{t,\mathbf{E}} \in \mathbb{R}^p$ is a mean-zero white noise, and $\mathbf{A}_{\mathbf{E}}$ has orthonormal columns.

Conversely, if (3) holds with (F') for some basis $\mathbf{E}$, then (2) holds with (F).

The factor representation (3) is basis-dependent. If a different set of orthonormal basis $\mathbf{D}$ is used, then there exists an unitary $\mathbf{Q} \in \mathbb{R}^{p \times p}$ such that $\mathbf{z}_{t,\mathbf{D}} = \mathbf{Q}\mathbf{z}_{t,\mathbf{E}} = \mathbf{Q}\mathbf{A}_{\mathbf{E}}\mathbf{f}_t + \mathbf{Q}\boldsymbol{\delta}_{t,\mathbf{E}}$, and

$\mathbf{z}_{t,\mathbf{D}}$ still admits a factor model structure by Proposition 1. In practice, we often need to use a specific basis for computation (see the next subsection), but it is easier to work with the intrinsic formulation (2), which avoids the ambiguity of the choice of the basis, in the theoretical expositions.

We make some assumptions about the RFM and discuss here their implications. First, as noted above, we assume $\boldsymbol{\delta}_t$ is a white noise while $\mathbf{f}_t$ drives all dynamics. Specifically, we assume $\{\mathbf{f}_t\}$ and $\{\boldsymbol{\delta}_t\}$ are weakly stationary processes such that $\mathbb{E}(\boldsymbol{\delta}_t \otimes \boldsymbol{\delta}_{t+h}) = 0$ and $\mathbb{E}(\boldsymbol{\delta}_{t+h} \otimes \mathbf{f}_t) = 0$ for $h \neq 0$, where for two vectors $\mathbf{u}, \mathbf{v}$ from the vector spaces U, V , respectively, the operator $\mathbf{u} \otimes \mathbf{v} : U \rightarrow V$ is defined by $(\mathbf{u} \otimes \mathbf{v})\mathbf{w} = \langle \mathbf{u}, \mathbf{w} \rangle \mathbf{v}$ for all $\mathbf{w} \in U$. Second, let Q_t be the probability measure of $\mathbf{x}_t$. The Fréchet mean of Q_t is defined as the minimizer of

$$F_t(\mathbf{y}) = \int d_M(\mathbf{y}, \mathbf{x})^2 Q_t(d\mathbf{x}), \quad \mathbf{y} \in M,$$

provided the minimizer exists and is unique. We assume the following about $\{\mathbf{x}_t\}$.

(P1) There exists $R > 0$ such that the geodesic ball around $\boldsymbol{\mu}$ with radius R , $B_{\boldsymbol{\mu}}(R) = \{\mathbf{x} \in M : d_M(\mathbf{x}, \boldsymbol{\mu}) < R\}$, is a normal neighborhood of $\boldsymbol{\mu}$, and there exists some small $\eta > 0$ such that $\mathbb{P}(\mathbf{x}_t \in B_{\boldsymbol{\mu}}(R/(2 + \eta))) = 1$ for all t .

(P2) The Fréchet mean of Q_t exists and is equal to $\boldsymbol{\mu}$ for all t .

(P1) assumes the data cluster around $\boldsymbol{\mu}$, which implies the geodesic joining $\mathbf{x}_t$ and $\boldsymbol{\mu}$ varies smoothly and that closed subsets in this neighborhood are compact. For spheres, $B_{\boldsymbol{\mu}}(R)$ is a normal neighborhood if $R \leq \pi$. For negatively curved manifolds, such as the SPD matrices equipped the affine-invariant metric, exponential maps are global diffeomorphisms on these manifolds. It follows that geodesic balls around $\boldsymbol{\mu}$ are totally normal neighborhoods, and (P1) holds as long as the data are within bounded distance from $\boldsymbol{\mu}$. By (1), (P1), and an argument similar to Theorem 2.1 of Bhattacharya & Patrangenaru (2003), the set of

minimizers of F_t over M is nonempty. Thus, for the existence of the Fréchet mean (P2) essentially assumes uniqueness of the minimizer. In general, uniqueness is guaranteed if Q_t is sufficiently concentrated, depending on the curvature of the underlying manifold (see Remark 2.1 of [Bhattacharya & Patrangenaru 2003](#)).

Remark 1. *The second part of (P2) requires the Fréchet means of $\mathbf{x}_t$, $t = 1, 2, \dots$, are equal to $\boldsymbol{\mu}$. From a modeling perspective, this assumption is quite natural. Indeed, if the Fréchet mean of Q_t is $\boldsymbol{\nu}$ and if $\mathbb{P}(\mathbf{x}_t \in B_{\boldsymbol{\nu}}(R_{\boldsymbol{\nu}})) = 1$, where $R_{\boldsymbol{\nu}}$ is the radius of injectivity at $\boldsymbol{\nu}$, then it follows from Theorem 2.1 of [Bhattacharya & Patrangenaru \(2003\)](#) that*

$$\int \text{Log}_{\boldsymbol{\nu}}(\mathbf{x}) Q_t(d\mathbf{x}) = 0,$$

where $\text{Log}_{\boldsymbol{\nu}} : B_{\boldsymbol{\nu}}(R_{\boldsymbol{\nu}}) \rightarrow T_{\boldsymbol{\nu}}M$ is the inverse of the exponential map at $\boldsymbol{\nu}$, restricted to the ball centered at zero with the radius $R_{\boldsymbol{\nu}}$. Then Eq. (2) simply postulates a factor model for the mean-zero process $\mathbf{z}_t = \text{Log}_{\boldsymbol{\nu}}(\mathbf{x}_t)$ in the tangent space.

In high-dimensional statistics, concentration inequalities are indispensable tools ([Wainwright 2019](#), [Bühlmann & van de Geer 2011](#), [Boucheron et al. 2013](#)). For RFM, we assume the following.

(P3) $\{\mathbf{z}_t\}$ satisfies the concentration inequality

$$\sup_{\mathbf{w} \in B_{\boldsymbol{\mu}}(R)} \left| \frac{1}{n} \sum_{t=1}^n [d_M(\mathbf{w}, \text{Exp}_{\boldsymbol{\mu}}(\mathbf{z}_t))^2 - \mathbb{E}(d_M(\mathbf{w}, \text{Exp}_{\boldsymbol{\mu}}(\mathbf{z}_t))^2)] \right| = O_p(c_n),$$

where $c_n \rightarrow 0$.

In general, (P3) is a special case of the concentration inequality

$$\sup_{f \in \mathcal{F}_n} \left| \frac{1}{n} \sum_{t=1}^n [f(\mathbf{z}_t) - \mathbb{E}f(\mathbf{z}_t)] \right| = O_p(c_n),$$

where $f : T_{\boldsymbol{\mu}}M \rightarrow \mathbb{R}$ and $\mathcal{F}_n$ is some suitable class of functions. Such concentration inequalities hold for a wide variety of time series processes. The concentration rate c_n typically hinges on the temporal dependence structure and the complexity of the function

class $\mathcal{F}_n$. For instance, [Chen & Wu \(2018\)](#) and [Deb & Mukherjee \(2024\)](#) established concentration inequalities under various dependence assumptions characterized by the rate of decay in moving average coefficients or mixing conditions, both of which allow long-range dependent and heavy-tailed processes. See also [Yu \(1994\)](#) for a uniform bound characterized by the metric entropy of $\mathcal{F}_n$ and the decay rate of mixing coefficients. Note that the dependence on the manifold dimension p is absorbed into the rate c_n . Below we give an example where c_n is explicitly calculated.

Example 1. Suppose [\(P1\)](#) holds and M is complete with sectional curvature bounded below by $K_1 \in \mathbb{R}$. Assume $\{\mathbf{z}_t\}$ is a strictly stationary α -mixing process with α -mixing coefficients $\{\alpha_j\}$. Then

- (Geometric decay) If $\{\alpha_k\}$ satisfies $\alpha_k \leq H\rho^k$ for some $H > 0$, $\rho \in (0, 1)$ and if $p = o(n^\gamma / \log n)$ for some $\gamma \in (0, 0.5)$, then [\(P3\)](#) holds with $c_n = n^{-(1-2\gamma)/2}$.
- (Algebraic decay) If $\alpha_k \leq Hk^{-\zeta}$ for some $H > 0$ and $\zeta > 0$, and if p is fixed, then [\(P3\)](#) holds with $c_n = n^{-\frac{\zeta}{3/2+p+2\zeta}} \log n$.

This example indicates a trade-off between the serial dependence in the data and the manifold dimension p . If $\mathbf{z}_t$ has geometrically decaying α -mixing coefficients, [\(P3\)](#) holds with p diverging at a rate slower than $\sqrt{n}$. If the α -mixing coefficients decay algebraically, a sufficient condition for [\(P3\)](#) is that p is fixed, with the convergence rate c_n adversely affected by a large p . The faster the α -mixing coefficients tend to zero (a large ζ), the faster the convergence rate c_n is, which is close to the ideal $n^{-1/2}$ rate in the limiting case where $\zeta \rightarrow \infty$. In [Section S.1](#) of the supplementary material, we also discuss the α -mixing coefficients of high-dimensional linear processes, which may be of independent interest. In particular, we show that for a high-dimensional VAR(1) process, the α -mixing coefficients decay geometrically but the constant H in [Example 1](#) can scale polynomially with p . Nevertheless, the result of [Example 1](#) still holds for such process because H only

has a logarithmic effect (see Example S.1 and Remark S.3 in the supplementary material).

For the manifold M , we assume

- (M1) The sectional curvature satisfies $K_1 \leq K(\mathbf{x}, \sigma) \leq K_0$ for all $\mathbf{x} \in B_\mu(R)$ and all two-dimensional subspace $\sigma \subset T_{\mathbf{x}}M$, for some $K_0 < \infty$ and $K_1 > -\infty$.

For many standard manifolds, either K_0 or K_1 is trivial. For example, the sphere or the Bures–Wasserstein manifold are nonnegatively curved, i.e., $K(\mathbf{x}, \sigma) \geq 0$, while the space of SPD matrices equipped with the affine-invariant metric satisfies $K(\mathbf{x}, \sigma) \leq 0$.

In closing the subsection, we remark that the Riemannian factor model can be employed for *multivariate* manifold-valued data. For instance, if $\mathbf{x}_t$ and $\mathbf{y}_t$ are time series taking values in the Riemannian manifolds M_1 and M_2 respectively, then one can view $(\mathbf{x}_t, \mathbf{y}_t)$ as a time series in the product manifold $M_1 \times M_2$, which inherits a Riemannian structure. In this setting, the RFM captures the lower-dimensional factors driving the joint dynamic dependence of $\mathbf{x}_t$ and $\mathbf{y}_t$. In Section 4, we investigate the application of RFM to the product manifold of spheres in the simulation study.

2.3 Estimation

Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n \in M$ be the observed time series. We estimate the RFM as follows. First, the Fréchet mean is estimated by optimizing

$$\hat{\boldsymbol{\mu}} \in \arg \min_{\mathbf{y} \in M} \sum_{t=1}^n d_M(\mathbf{y}, \mathbf{x}_t)^2.$$

Numerically, Riemannian gradient methods are often employed to solve this minimization problem (see, for example, Fletcher et al. 2004). Let $\hat{\mathbf{E}} = (\hat{\mathbf{e}}_1, \dots, \hat{\mathbf{e}}_p)$ be an orthonormal basis of $T_{\hat{\boldsymbol{\mu}}}M$. Then we can express the log-mapped data $\hat{\mathbf{z}}_t = \text{Log}_{\hat{\boldsymbol{\mu}}}(\mathbf{x}_t)$ with respect to $\hat{\mathbf{E}}$, denoted as $\hat{\mathbf{z}}_{t, \hat{\mathbf{E}}}$. Let $\hat{\mathbf{S}}_{\hat{\mathbf{E}}}(h) = n^{-1} \sum_{t=h_0+1}^n \hat{\mathbf{z}}_{t, \hat{\mathbf{E}}} \hat{\mathbf{z}}_{t-h, \hat{\mathbf{E}}}^\top$, $h = 1, 2, \dots, h_0$, be the sample autocovariance matrices expressed in $\hat{\mathbf{E}}$, where $h_0 \geq 1$ is a prescribed positive integer. Following Lam

et al. (2011), put $\hat{\mathbf{L}}_{\hat{\mathbf{E}}} = \sum_{h=1}^{h_0} \hat{\mathbf{S}}_{\hat{\mathbf{E}}}(h) \hat{\mathbf{S}}_{\hat{\mathbf{E}}}(h)^\top$ and compute the first r eigenvectors $\hat{\mathbf{A}}_{\hat{\mathbf{E}}} = (\hat{\mathbf{a}}_{1,\hat{\mathbf{E}}}, \dots, \hat{\mathbf{a}}_{r,\hat{\mathbf{E}}})$ corresponding the largest r eigenvalues of $\hat{\mathbf{L}}_{\hat{\mathbf{E}}}$. Subsequently, the factor process can be estimated by $\hat{\mathbf{f}}_t = \hat{\mathbf{A}}_{\hat{\mathbf{E}}}^\top \hat{\mathbf{z}}_{t,\hat{\mathbf{E}}}$. Let $J_{\hat{\mathbf{E}}} : \mathbb{R}^p \rightarrow T_{\hat{\boldsymbol{\mu}}}M$ be defined as

$$J_{\hat{\mathbf{E}}}(a_1, \dots, a_p) = \sum_{j=1}^p a_j \hat{\mathbf{e}}_j. \quad (4)$$

That is, $J_{\hat{\mathbf{E}}}$ identifies the $\hat{\mathbf{E}}$ -coordinate vector with the corresponding element in the tangent space. Then the (in-sample) model predictions are $\hat{\mathbf{x}}_t = \text{Exp}_{\hat{\boldsymbol{\mu}}}(J_{\hat{\mathbf{E}}}(\hat{\mathbf{A}}_{\hat{\mathbf{E}}} \hat{\mathbf{f}}_t))$. In practice, the number of factors r is unknown. We may estimate the number of factors by

$$\hat{r} = \arg \min_{1 \leq i \leq R} \hat{\lambda}_{i+1} / \hat{\lambda}_i, \quad (5)$$

where $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \dots$ are the eigenvalues of $\hat{\mathbf{L}}_{\hat{\mathbf{E}}}$. For Euclidean data, this ratio estimator is employed by Lam & Yao (2012).

Due to the intrinsic formulation of the Riemannian factor model, the evaluation of the accuracy and performance requires special consideration. Our focus is on the uniquely defined loading space in the factor model (2), which is the image space of $\mathcal{A}$, $\text{Im}(\mathcal{A})$. To avoid the dependence on the choice of the basis, we shall refer some key quantities above to their intrinsic counterparts. Specifically, for $\hat{\mathbf{S}}_{\hat{\mathbf{E}}}$ and $\hat{\mathbf{L}}_{\hat{\mathbf{E}}}$, their intrinsic, basis-independent versions are, respectively,

$$\hat{\mathcal{S}}(h) = \frac{1}{n} \sum_{t=h_0+1}^n \hat{\mathbf{z}}_{t-h} \otimes \hat{\mathbf{z}}_t, \text{ and } \hat{\mathcal{L}} = \sum_{h=1}^{h_0} \hat{\mathcal{S}}(h) \hat{\mathcal{S}}(h)^*,$$

where $\mathcal{A}^*$ denotes the adjoint of the operator $\mathcal{A}$. Let $\hat{\mathbf{a}}_1, \dots, \hat{\mathbf{a}}_r \in T_{\hat{\boldsymbol{\mu}}}M$ be the leading r eigenvectors of $\hat{\mathcal{L}}$, which spans the estimated loading space $\text{span}\{\hat{\mathbf{a}}_1, \dots, \hat{\mathbf{a}}_r\}$. Note that $\{\hat{\lambda}_i : i = 1, 2, \dots\}$ in Eq. (5) are also eigenvalues of $\hat{\mathcal{L}}$.

Note that $\hat{\mathcal{S}}(h)$, $\hat{\mathcal{L}}$, and $\{\hat{\mathbf{a}}_j\}$ are associated with $T_{\hat{\boldsymbol{\mu}}}M$, which are not directly comparable to the counterparts acting on $T_{\boldsymbol{\mu}}M$. To tackle this issue in theoretical investigation and in simulation studies, we parallel transport the estimated loading space to a subspace in

$T_{\mu}M$ along the unique geodesic between $\hat{\mu}$ and μ , and compare the two subspaces using the sine-theta distance. In particular, we measure

$$\|\sin \Theta(\text{span}\{\hat{\mathcal{P}}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_1, \dots, \hat{\mathcal{P}}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_r\}, \text{Im}(\mathcal{A}))\|_{op}, \quad (6)$$

where $\sin \Theta(\mathcal{W}, \mathcal{V})$ is the diagonal matrix of the sine-theta angles between the two linear spaces $\mathcal{W}, \mathcal{V} \subset T_{\mu}M$ (Davis & Kahan 1970, Chen et al. 2021), and $\|\cdot\|_{op}$ denotes the operator norm. This strategy, also employed by Lin & Yao (2019), involves only intrinsic quantities and therefore can be evaluated without recourse to an ambient space in which the manifold M is embedded. For manifolds like the Bures–Wasserstein manifold which lacks a canonical ambient space, this is particularly useful.

Remark 2. Following Lin & Yao (2019), a linear operator $\hat{\mathcal{B}} : T_{\hat{\mu}}M \rightarrow T_{\hat{\mu}}M$ can be compared with the linear operator $\mathcal{B} : T_{\mu}M \rightarrow T_{\mu}M$ through the parallel transporter $\Phi_{\hat{\mu}, \mu}$, defined as

$$(\Phi_{\hat{\mu}, \mu} \hat{\mathcal{B}}) \mathbf{v} = \mathcal{P}_{\hat{\mu} \rightarrow \mu} \hat{\mathcal{B}}(\mathcal{P}_{\mu \rightarrow \hat{\mu}} \mathbf{v}) \quad \text{for all } \mathbf{v} \in T_{\mu}M. \quad (7)$$

Let $\mathcal{B} = \mathcal{A}\mathcal{A}^*$ and $\hat{\mathcal{B}} = \sum_{j=1}^r \hat{\mathbf{a}}_j \otimes \hat{\mathbf{a}}_j$. Since $\text{Im}(\mathcal{B}) = \text{Im}(\mathcal{A})$ and $\text{Im}(\Phi_{\hat{\mu}, \mu} \hat{\mathcal{B}}) = \text{span}\{\mathcal{P}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_1, \dots, \mathcal{P}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_r\}$, the sine-theta distance

$$\|\sin \Theta(\text{span}\{\hat{\mathcal{P}}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_1, \dots, \hat{\mathcal{P}}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_r\}, \text{Im}(\mathcal{A}))\|_{op}$$

can be interpreted as the discrepancy between the operators $\mathcal{B}$ and $\hat{\mathcal{B}}$ under parallel transporter.

3 Theoretical results

In this section, we present the theoretical properties of the proposed estimators. We first establish a convergence rate of the estimators under fairly mild assumptions. Then, we show that with a short-memory and a strong factor assumption, a dimension-free convergence rate can be attained. All proofs are relegated to the supplementary material.

To begin, we show that the estimator $\hat{\boldsymbol{\mu}}$ for the Fréchet mean $\boldsymbol{\mu}$ is consistent. For this, we assume the following identification conditions.

(P4) For any $\epsilon > 0$,

$$\inf_{n \in \mathbb{N}} \inf_{t \in [n]} \inf_{\mathbf{w}: d_M(\mathbf{w}, \boldsymbol{\mu}) > \epsilon} \{ \mathbb{E}(d_M(\mathbf{w}, \mathbf{x}_t)^2) - \mathbb{E}(d_M(\boldsymbol{\mu}, \mathbf{x}_t)^2) \} > 0.$$

(P5) Let $g_n(\mathbf{v}) = n^{-1} \sum_{t=1}^n \int d_M(\text{Exp}_{\boldsymbol{\mu}}(\mathbf{v}), \mathbf{x})^2 Q_t(d\mathbf{x})$. There exists some $B > 0$ such that

$$\sup_{\|\mathbf{v}\| < B} \lambda_{\min}^{-1} \left(\frac{\partial^2}{\partial \mathbf{v}^2} g_n(\mathbf{v}) \right) = O(1).$$

(P4) and (P5) assume the loss function $n^{-1} \sum_{t=1}^n \int d_M(\cdot, \mathbf{x})^2 Q_t(d\mathbf{x})$ is well-behaved around the minimizer $\boldsymbol{\mu}$. Similar assumptions were also adopted in Dai & Müller (2018) and Lin & Yao (2019).

Proposition 2. Assume (P1)–(P4). Then

$$d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) = o_p(1). \quad (8)$$

Moreover, if (P1)–(P5) holds, then

$$d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) = O_p(c_n^{1/2}), \quad (9)$$

where c_n is given in assumption (P3).

With the convergence rate of the sample Fréchet mean $\hat{\boldsymbol{\mu}}$ to the Fréchet mean $\boldsymbol{\mu}$ in Proposition 2, next we establish a convergence rate for the estimated loading space, which is given in Theorem 1 below.

Theorem 1. Assume (M1), (P1)–(P5), and that there exists $\delta \in (0, \eta)$ with η defined in Assumption (P1) such that the radius of injectivity of every point of $B_{\boldsymbol{\mu}}(R/(2 + \eta))$ is at least $(1 + \delta)R/(2 + \eta)$. If $R < (\frac{2+\eta}{1+\delta}) \frac{\pi}{\sqrt{K_0}}$ with K_0 given in Assumption (M1), then

$$\|\Phi_{\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}} \hat{\mathcal{L}} - \mathcal{L}\|_{op} = O_p(c_n^{1/2}), \quad (10)$$

where $\Phi_{\hat{\mu}, \mu}$ is defined in Eq. (7), $\mathcal{L} = \sum_{h=1}^{h_0} \mathcal{S}(h) \mathcal{S}(h)^*$ and $\mathcal{S}(h) = n^{-1} \sum_{t=h_0+1}^n \mathbf{z}_{t-h} \otimes \mathbf{z}_t$, and c_n is as in Assumption (P3). In addition, if for any fixed positive integer h ,

$$\left\| \frac{1}{n} \sum_{t=h+1}^n (\mathbf{z}_t \otimes \mathbf{z}_{t-h} - \mathbb{E}(\mathbf{z}_t \otimes \mathbf{z}_{t-h})) \right\|_{\text{HS}} = O_p(\epsilon_n), \quad (11)$$

where $\epsilon_n = o(1)$ and $\|\cdot\|_{\text{HS}}$ denotes the Hilbert–Schmidt norm, then the space spanned by the first r eigenvectors, $\hat{\mathbf{a}}_1, \dots, \hat{\mathbf{a}}_r$, of $\hat{\mathcal{L}}$ satisfies

$$\|\sin \Theta(\text{span}\{\mathcal{P}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_1 \dots \mathcal{P}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_r\}, \text{Im}(\mathcal{A}))\|_{op} = O_p\left(\frac{\sqrt{c_n} + \epsilon_n}{\kappa^2}\right), \quad (12)$$

where r is the dimension of the factor process $\mathbf{f}_t$ as defined after Eq. (2), $\sin \Theta(\cdot, \cdot)$ is defined in Eq. (6), and $\kappa = \max_{1 \leq h \leq h_0} \sigma_r(\mathbb{E}(\mathbf{f}_{t-h} \otimes \mathbf{f}_t))$ with $\sigma_k(\cdot)$ denoting the k -th singular value (in descending order).

Theorem 1 is novel in two aspects. First, unlike the convergence rates obtained in Dai & Müller (2018) and Lin & Yao (2019) for the Riemannian functional PCA, we explicitly track the dependence of the convergence rate on the manifold’s dimension p , absorbed in c_n and ϵ_n , instead of treating it as fixed. This distinction is crucial, as dimensionality often plays an important role in the convergence rates for the high-dimensional factor models. In this regard, our result is reminiscent of the high-dimensional linear factor models for Euclidean time series, for which it was shown that $\kappa^2 = O(p)$ and $\epsilon_n = O(p/\sqrt{n})$, assuming strong factors in the sense of Lam et al. (2011). Second, the manifold is not assumed to be complete. In particular, this makes Theorem 1 applicable to time series in the Bures–Wasserstein manifold.

The theoretical analysis requires bounding $\mathcal{P}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{z}}_t - \mathbf{z}_t$ non-asymptotically with careful geometric comparisons. Standard tools used in the existing statistical analysis of manifold-valued data (Bhattacharya & Patrangenaru 2005, 2003, Kendall & Le 2011, Lin & Yao 2019), such as the covariant Taylor expansion of the log map, are inadequate here. To overcome these challenges, we employ a holonomy argument that tracks changes in tangent

vectors parallelly transported along small geodesic triangles. To the best of our knowledge, this technique has not appeared in previous works.

Assumption (P3) and Eq. (11) in Theorem 1 are very weak assumptions on the serial dependence in the data, as they permit long-range dependent time series. Theorem 2 below refines Theorem 1 and obtains a faster rate of convergence under an additional short-memory assumption. Before presenting the result, we need to introduce some notations. For each $\xi \in B_\mu(R/(2+\eta))$, let $\mathbf{z}_{t,\xi} = \text{Log}_\xi(\mathbf{x}_t)$. The autocovariance operators of the process $\{\mathbf{z}_{t,\xi}\}$ is denoted by

$$\Gamma_\xi(s, t) = \mathbb{E}[(\mathbf{z}_{s,\xi} - \mathbb{E}\mathbf{z}_{s,\xi}) \otimes (\mathbf{z}_{t,\xi} - \mathbb{E}\mathbf{z}_{t,\xi})].$$

Now we can state Theorem 2.

Theorem 2. *Assume the same as in Theorem 1. Let $\mathbf{E} = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_p)$ be an orthonormal basis of $T_\mu M$. Suppose each component of $\mathbf{z}_{t,\mathbf{E}}$, $z_{t,j} = \langle \mathbf{z}_t, \mathbf{e}_j \rangle$, $j = 1, 2, \dots, p$, admits an $MA(\infty)$ representation $z_{t,j} = \sum_{k=0}^{\infty} \alpha_{k,j} \epsilon_{t-k,j}$, where $\max_{1 \leq j \leq p} \sum_k \alpha_{k,j}^2 = O(1)$ and $\{\epsilon_{t,j}\}$ is an i.i.d. sequence of random variables such that*

$$\sup_{\substack{j \in \{1, 2, \dots, p\} \\ -\infty < t < \infty}} \mathbb{E}(|\epsilon_{t,j}|^4) = O(1).$$

In addition, there exists some $\epsilon > 0$ with the property that for each $\xi \in B_\mu(\epsilon)$, there exist $C_\xi < \infty$ and $d_\xi > 1$ such that

$$\frac{|\text{Tr}(\Gamma_\xi(s, t))|}{\sqrt{\text{Tr}(\Gamma_\xi(s, s))\text{Tr}(\Gamma_\xi(t, t))}} \leq C_\xi |s - t|^{-d_\xi} \quad (13)$$

for all s, t , where

$$\sup_{\xi \in B_\mu(\epsilon)} C_\xi = O(1), \text{ and } \inf_n \inf_{\xi \in B_\mu(\epsilon)} d_\xi > 1. \quad (14)$$

Then

$$\|\sin \Theta(\text{span}\{\mathcal{P}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_1 \dots \mathcal{P}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_r\}, \text{Im}(\mathcal{A}))\|_{op} = O_p\left(\frac{1}{\kappa^2 \sqrt{n}}\right) \quad (15)$$

where κ is as in Theorem 1 and $\sin \Theta(\cdot, \cdot)$ is defined after Eq. (6).

As shown in Theorem 2, under the short-memory condition (13)–(14) the estimated loading space converges at the dimension-free $n^{-1/2}$ rate if $\kappa \asymp 1$. Thus, the Riemannian factor model can be estimated as efficiently as the high-dimensional linear factor model with strong factors (Lam et al. 2011) and the fixed-dimensional Riemannian functional PCA (Lin & Yao 2019, Dai & Müller 2018). We therefore refer to the case $\kappa \asymp 1$ as “strong factors,” and the case $\kappa = o(1)$ as “weak factors” for the Riemannian factor model.

It is worth pointing out that the dimension-free rate is not a straightforward extension from those earlier results. In fact, both the signal strength and the estimation noise are quite different from the classical linear factor model. Recall that in Theorem 1, the convergence rate for the estimated loading space is determined by the signal-to-noise ratio, in which the estimation noise is quantified by $\sqrt{c_n} + \epsilon_n$, and the signal strength by κ^2 . For the signal strength, (P1) implies

$$\mathbb{E}\|\mathbf{z}_t\|_{\boldsymbol{\mu}}^2 = \mathbb{E}\|\mathcal{A}\mathbf{f}_t\|_{\boldsymbol{\mu}}^2 + \mathbb{E}\|\boldsymbol{\delta}_t\|_{\boldsymbol{\mu}}^2 = \mathbb{E}\|\mathbf{f}_t\|^2 + \mathbb{E}\|\boldsymbol{\delta}_t\|_{\boldsymbol{\mu}}^2 \leq \left(\frac{R}{2+\eta}\right)^2.$$

In particular, neither $\mathbb{E}\|\mathbf{f}_t\|^2$ nor $\mathbb{E}\|\boldsymbol{\delta}_t\|_{\boldsymbol{\mu}}^2$ can diverge with the dimension p , so the signal strength κ^2 is of order $O(1)$. For the noise component in Theorem 1, the term $c_n^{1/2}$ accounts for the estimation error of the Fréchet mean $d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu})$, while the term ϵ_n reflects the estimation error of the autocovariance operators. Both errors, under the additional assumptions in Theorem 2, are of order $n^{-1/2}$, independent of p .

The additional assumptions in Theorem 2 are rather mild. First, the i.i.d. assumption on $\{\epsilon_{t,j}\}$ can be relaxed to allow for martingale difference sequences with conditional heteroscedasticity, at the expense of stronger moment conditions (see, e.g., Theorem A.2 in Huang et al. 2025). Second, except when $\boldsymbol{\xi} = \boldsymbol{\mu}$, the process $\mathbf{z}_{t,\boldsymbol{\xi}}$ may not be weakly stationary. Clearly, $\mathbf{z}_{t,\boldsymbol{\xi}}$ is stationary if $\{\mathbf{z}_t\}$ is strictly stationary, in which case $\boldsymbol{\Gamma}_{\boldsymbol{\xi}}(s, t)$ depends only on $|s - t|$. Finally, condition (13), requiring the autocovariances of $\mathbf{z}_{t,\boldsymbol{\xi}}$ to decay sufficiently fast, ensures short-memory. It is natural in our setting because the

autocovariance operator $\mathbf{\Gamma}_{\boldsymbol{\xi}}$ varies differentiably near $\boldsymbol{\mu}$. Moreover, it automatically holds when $\{\mathbf{z}_t\}$ is m -dependent for some finite m . In that case, since $\mathbf{x}_t$ and $\mathbf{x}_{t-k}$ are independent for $k > m$, $\mathbf{\Gamma}_{\boldsymbol{\xi}}(s, t) = 0$ for all $|s - t| > m$.

Our final result, which gives convergence rates for the estimated eigenvalues, justifies the ratio estimator in Eq. (5) for selecting the number of factors. These rates also coincide with the results of Lam & Yao (2012) for the Euclidean time series.

Proposition 3. *Under the assumptions of Theorem 2, we have*

$$(i) \quad |\hat{\lambda}_i - \lambda_i| = O_p(n^{-1/2}) \text{ for } i = 1, 2, \dots, r;$$

$$(ii) \quad \hat{\lambda}_i = O_p(n^{-1}) \text{ for } i > r,$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$ are the eigenvalues of the operator $\mathbf{\Lambda} = \sum_{h=1}^{h_0} \mathbb{E}(\mathbf{z}_{t-h} \otimes \mathbf{z}_t) \mathbb{E}(\mathbf{z}_{t-h} \otimes \mathbf{z}_t)^*$. In addition, $\lambda_r \geq \kappa^2$ and $\lambda_i = 0$ for $i > r$.

4 Simulation studies

In this section, we assess the performance of the proposed RFM using synthetic data. We consider data situated on two nonlinear manifolds, namely the Bures–Wasserstein manifold and the product manifold of spheres. Due to space constraints, we present the results for the Bures–Wasserstein manifold in Section S.3 of the supplementary material. To illustrate the gains in capturing the underlying nonlinear geometry, we compare the RFM with the linear factor model (LFM) in terms of the pseudo-prediction errors. The results show that the RFM can achieve a low pseudo-prediction error with a few factors. In addition, when the sample size is sufficiently large, the selected number of factors and the estimated loading space are also accurate, supporting the theoretical results.

In the following, we consider sphere-valued time series. Specifically, we apply the proposed

RFM to the more complex setup where the time series take values in the product of unit spheres $M = \mathbb{S}^{q_1} \times \dots \times \mathbb{S}^{q_d}$, equipped with the product metric, where $\mathbb{S}^q$ denotes the unit sphere in $\mathbb{R}^{q+1}$. Sphere-valued data are commonly seen in environmental studies, such as wind directions or ocean currents, as well as in statistical shape analysis (Dryden & Mardia 2016). They have also attracted much attention due to its connection to compositional data (Wang et al. 2007, Dai & Müller 2018).

The synthetic data are generated as follows. The reference point $\boldsymbol{\mu} = (\boldsymbol{\mu}_1, \dots, \boldsymbol{\mu}_d) \in M$ has each of its components $\boldsymbol{\mu}_j \in \mathbb{S}^{q_j}$ uniformly selected on its corresponding sphere. Let $\mathbf{f}_t \in \mathbb{R}^r$ be the factor process, which follows an AR(1) model

$$\mathbf{f}_t = \alpha \mathbf{f}_{t-1} + s \cdot \boldsymbol{\epsilon}_t, \quad (16)$$

where $|\alpha| < 1$, $s > 0$ and $\boldsymbol{\epsilon}_t = (\epsilon_{t,1}, \dots, \epsilon_{t,r})^\top$ is drawn independently from the standard normal distribution truncated at the unit ball to ensure bounded support. In this subsection, we set $s = 2.5\sqrt{d/5}$ and $\alpha = 0.5$. Let $\mathbf{E} = (\mathbf{e}_1, \dots, \mathbf{e}_p)$ be an orthonormal basis of $T_{\boldsymbol{\mu}}M$, and $\mathbf{A}_{\mathbf{E}} \in \mathbb{R}^{p \times r}$, where $p = q_1 + \dots + q_d$ is the manifold dimension, be a matrix with randomly generated orthonormal columns. In addition, $\boldsymbol{\delta}_{t,\mathbf{E}}$ is independently generated from the uniform distribution on the p -sphere of radius σ_z and set $\mathbf{z}_{t,\mathbf{E}} = \mathbf{A}_{\mathbf{E}}\mathbf{f}_t + \boldsymbol{\delta}_{t,\mathbf{E}}$. Finally, the observed data $\mathbf{x}_t$ is obtained through the exponential map $\mathbf{x}_t = \text{Exp}_{\boldsymbol{\mu}}(\mathbf{z}_t)$, where $\mathbf{z}_t = J_{\mathbf{E}}(\mathbf{z}_{t,\mathbf{E}})$ with $J_{\mathbf{E}}$ defined in (4). We consider the cases $d \in \{5, 10, 20\}$ and set $q_j = 4$ for $j = 1, 2, \dots, d$.

To illustrate the impact of the nonlinear constraint of the manifold, we compare the RFM with the linear factor model (LFM) of Lam et al. (2011). In practice, it is not uncommon to treat $\mathbf{x}_{t,j}$ as Euclidean data in the usual coordinates of $\mathbb{R}^{q_j+1}$, denoted hereafter by $\mathcal{E}(\mathbf{x}_{t,j}) \in \mathbb{R}^{q_j+1}$. Hence, in the following experiments, we apply the LFM to $\mathcal{E}(\mathbf{x}_t) = (\mathcal{E}(\mathbf{x}_{t,1}), \dots, \mathcal{E}(\mathbf{x}_{t,d})) \in \mathbb{R}^{\sum_{j=1}^d (q_j+1)}$.

For each experiment, a test set of size $n_{\text{test}} = 200$ is reserved to evaluate the performance.

Let $\hat{\mathbf{x}}_t^{(L)}$ and $\hat{\mathbf{x}}_t^{(R)}$ be the pseudo-predictions of LFM and RFM, respectively. Specifically, at time t , given a new observation $\mathbf{x}_t$, the pseudo-prediction of RFM is given by $\hat{\mathbf{x}}_t^{(R)} = \text{Exp}_{\hat{\boldsymbol{\mu}}}(J_{\hat{\mathbf{E}}}(\hat{\mathbf{A}}_{\hat{\mathbf{E}}} \hat{\mathbf{A}}_{\hat{\mathbf{E}}}^\top \hat{\mathbf{z}}_{t,\hat{\mathbf{E}}}))$, where $\hat{\boldsymbol{\mu}}$ is the estimated Fréchet mean, $\hat{\mathbf{E}}$ is an orthonormal basis of $T_{\hat{\boldsymbol{\mu}}}M$, $\hat{\mathbf{z}}_{t,\hat{\mathbf{E}}} = J_{\hat{\mathbf{E}}}^{-1}(\text{Log}_{\hat{\boldsymbol{\mu}}}(\mathbf{x}_t))$, and $\hat{\mathbf{A}}_{\hat{\mathbf{E}}}$ is the estimated loading matrix using the training sample $\mathbf{x}_1, \dots, \mathbf{x}_n$. Similarly, the pseudo-predictions for LFM are computed as $\mathcal{E}(\tilde{\mathbf{x}}_t^{(L)}) = \tilde{\boldsymbol{\mu}} + \tilde{\mathbf{A}}\tilde{\mathbf{A}}^\top \tilde{\mathbf{z}}_t$, where $\tilde{\mathbf{z}}_t = \mathcal{E}(\mathbf{x}_t) - \tilde{\boldsymbol{\mu}}$, $\tilde{\boldsymbol{\mu}}$ is the Euclidean mean of $\mathcal{E}(\mathbf{x}_1), \dots, \mathcal{E}(\mathbf{x}_n)$, and $\tilde{\mathbf{A}} \in \mathbb{R}^{p \times k}$ is the loading matrix estimated by LFM with k being the number of extracted factors. Note that $\mathcal{E}(\tilde{\mathbf{x}}_t^{(L)})$, the predictions of LFM, is not on the sphere in general. Thus, $\mathcal{E}(\hat{\mathbf{x}}_t^{(L)})$ is obtained by projecting $\mathcal{E}(\tilde{\mathbf{x}}_t^{(L)})$ to the unit spheres. Then, the relative geodesic mean squared error (rGMSE), defined as

$$\text{rGMSE} = \frac{\sum_{t=n+1}^{n+n_{\text{test}}} d_M(\mathbf{x}_t, \hat{\mathbf{x}}_t)^2}{\sum_{t=n+1}^{n+n_{\text{test}}} d_M(\mathbf{x}_t, \hat{\boldsymbol{\mu}})^2}, \quad (17)$$

is used to evaluate the methods.

As shown in Table 1, LFM and RFM achieve similar rGMSEs when fewer than 5 factors are used. However, when $r = 5$ factors are employed, RFM outperforms LFM, reducing the rGMSE by at least 20% compared to the LFM. In addition, to achieve the same level of rGMSE as the RFM with r factors, the LFM requires four or more additional factors. This highlights the effect of the manifold's nonlinearity on LFM, which cannot be easily compensated by simply increasing the number of factors. In addition, Table 2 shows the frequency with which the ratio estimator correctly identifies the number of factors. The correct number of factors is selected with high frequency—exceeding 80% of the time when $n = 100$ and near 100% when $n = 200$ in all cases, validating the efficacy of the ratio estimator in selecting the number of factors for the RFM.

Table 1: Relative GMSE of LFM and RFM applied to the product of spheres. The results are averaged over 300 simulations. The true number of factors $r = 5$.

Num. of factors	1	2	3	4	5	6	7	8	9	10
$(n, d) = (50, 5)$										
LFM	0.78	0.61	0.46	0.33	0.23	0.19	0.16	0.14	0.13	0.12
RFM	0.78	0.60	0.43	0.28	0.15	0.13	0.12	0.11	0.10	0.09
$(n, d) = (100, 5)$										
LFM	0.78	0.60	0.44	0.31	0.21	0.17	0.14	0.13	0.11	0.10
RFM	0.78	0.59	0.42	0.27	0.13	0.12	0.11	0.10	0.09	0.08
$(n, d) = (200, 5)$										
LFM	0.77	0.59	0.44	0.30	0.19	0.16	0.14	0.12	0.11	0.10
RFM	0.78	0.58	0.41	0.26	0.12	0.11	0.10	0.09	0.08	0.07
$(n, d) = (50, 10)$										
LFM	0.79	0.62	0.47	0.35	0.25	0.22	0.20	0.19	0.18	0.17
RFM	0.79	0.61	0.45	0.31	0.18	0.17	0.16	0.15	0.15	0.14
$(n, d) = (100, 10)$										
LFM	0.78	0.61	0.46	0.33	0.23	0.19	0.18	0.17	0.16	0.15
RFM	0.79	0.61	0.44	0.29	0.16	0.15	0.14	0.13	0.13	0.12
$(n, d) = (200, 10)$										
LFM	0.78	0.60	0.45	0.32	0.21	0.18	0.17	0.16	0.15	0.14
RFM	0.79	0.60	0.43	0.28	0.14	0.14	0.13	0.13	0.12	0.12
$(n, d) = (50, 20)$										
LFM	0.79	0.62	0.48	0.36	0.26	0.23	0.22	0.21	0.20	0.19
RFM	0.80	0.62	0.47	0.32	0.20	0.19	0.18	0.18	0.17	0.17
$(n, d) = (100, 20)$										
LFM	0.79	0.61	0.47	0.34	0.24	0.21	0.20	0.19	0.18	0.18
RFM	0.80	0.61	0.45	0.30	0.17	0.16	0.16	0.16	0.15	0.15
$(n, d) = (200, 20)$										
LFM	0.79	0.61	0.46	0.33	0.22	0.20	0.19	0.18	0.17	0.17
RFM	0.80	0.61	0.44	0.29	0.16	0.15	0.15	0.15	0.14	0.14

5 Empirical application

In this section, we apply the proposed RFM to U.S. financial data. In particular, we view the time-varying covariances (realized volatility) of the daily log returns of a selected set of U.S. stocks as a time series in the Bures–Wasserstein manifold. In addition to modeling the time series, we also investigate the out-of-sample forecasting performance of the RFM, which has important practical implications. The data used in the analysis were obtained from Yahoo! Finance, and the codes for data acquisition and analysis are available in the

Table 2: Proportion of times the correct number of factors is selected over 300 simulations. The data are on the product sphere $M = \mathbb{S}^{q_1} \times \dots \times \mathbb{S}^{q_d}$ generated according to Section 4.

$n \backslash d$	5	10	20
50	0.57	0.61	0.60
100	0.83	0.88	0.91
200	0.99	0.98	0.99

Table 3: The companies and their tickers used in the real data analysis.

Company	Ticker	Company	Ticker	Company	Ticker
Microsoft	MSFT	JPMorgan Chase	JPM	ExxonMobil	XOM
Apple Inc.	AAPL	Bank of America	BAC	Chevron Corporation	CVX
Oracle Corporation	ORCL	Wells Fargo	WFC	ConocoPhillips	COP
Cisco	CSCO	Goldman Sachs	GS	EOG Resources	EOG

Github repository [Omitted for anonymity].

We consider the daily log returns of the S&P 500 component stocks during January 1, 2000 to December 31, 2019, and retain only the companies with complete trading records in this period. Among them, we focus on the four largest companies by market capitalization from each of the three sectors: information technology (IT), finance, and energy. To keep the selection more homogeneous, we exclude Nvidia and Berkshire Hathaway, whose market dynamics differ from their corresponding sectors in ways that would complicate comparisons. The 12 selected companies, along with their tickers we used throughout the section, are listed in Table 3. For each month, we compute the realized covariance matrix $\Sigma_t \in \mathbb{R}^{12 \times 12}$ of the daily log returns (in percentage points), yielding 240 monthly matrix observations. The first 204 months are used as the training set and the final 36 months are reserved as test data.

First, we estimate the RFM and LFM using the training set and construct the pseudo-predictions as in Section 4 (for LFM applied to SPD matrices, see Section S.3 of the supplementary material). Table 4 reports the relative geodesic mean squared errors (rGMSE)

Table 4: Relative geodesic mean squared errors (rGMSE) of LFM and RFM applied to U.S. return data.

Num. of factors	1	2	3	4	5	6	7	8	9	10
LFM	1.183	0.533	0.455	0.429	0.423	0.410	0.406	0.410	0.405	0.391
RFM	0.539	0.404	0.344	0.323	0.295	0.267	0.258	0.244	0.238	0.229

defined in (17). The LFM attains an rGMSE of 0.405 using 9 factors. In contrast, the geometry-aware RFM achieves an rGMSE of 0.404 with only two factors. Figure 1 plots the leading factors estimated by the RFM. The first leading factor shows a pronounced spike for the 2008 financial crisis. The second factor has relatively larger values during the dot-com bubble, and it moves in the opposite direction during the financial crisis. To visualize and interpret the effects of these factors in the manifold, one can examine the geodesic trajectories traveled in the direction of the corresponding loading vectors. Specifically, given the estimated factor loading $\hat{\mathbf{a}}_j$ in the tangent space, the geodesic trajectory $\hat{\Sigma}_{(j)}(s) = \text{Exp}_{\hat{\Sigma}}(s\hat{\mathbf{a}}_j)$, $s \in \mathbb{R}$, where $\hat{\Sigma}$ is the estimated Fréchet mean, shows the effect of the j -th factor on the prediction, holding all other factors at zero. Figure 2 presents the resulting trajectories as heatmaps of the entries of $\hat{\Sigma}_{(j)}(s)$. The first factor primarily reflects the intensity of the interdependence between the selected companies, especially the financial companies. During crises, return correlations tend to be more elevated, and this factor captures such broad market-wide effect. Indeed, as shown in Figure 1, the first factor closely tracks the closing price of the VIX volatility index, a widely-used proxy for aggregate market uncertainty. Meanwhile, the second factor captures sector-specific volatility dynamics. As shown in Figure 2(b), positive values of the second factor, such as during the dot-com bubble, indicate heightened dependence within the IT sector. Negative values, such as during the financial crisis, shift the loading concentration to the financial and energy sectors.

Next, we evaluate the out-of-sample forecasting performance. We fit a VAR(1) model to

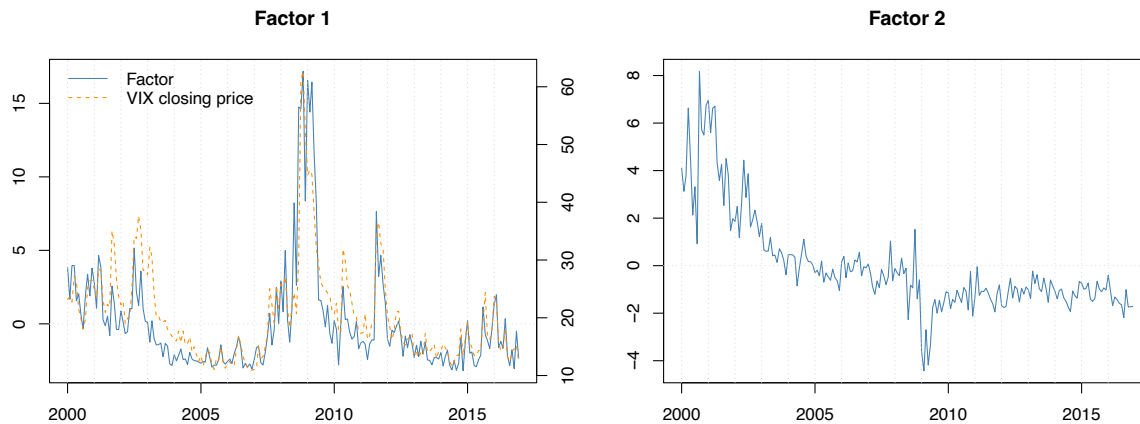


Figure 1: The leading two estimated factors (blue lines) from the U.S. financial data. The orange dashed line indicates the monthly average of VIX closing price.

the estimated factors using data up until time t , and obtain the one-step-ahead forecast $\hat{\mathbf{f}}_{t+1}$ and $\hat{\Sigma}_{t+1}$. The number of factors used in forecasting is selected by the ratio estimator in Eq. (5). As additional benchmarks, we employ two simple and widely used methods. The first is the last observation carried forward (LOCF) method, which uses the previous month’s covariance matrix as the forecast. The second is the exponentially weighted moving average (EWMA) method, implemented in RiskMetrics (Longerstaey & Zangari 1996; see also Johansson et al. 2023) developed by J. P. Morgan. Specifically, we implement the EWMA by the recursive formula $\hat{\Sigma}_{t+1} = \lambda \hat{\Sigma}_t + (1 - \lambda) \Sigma_t$, where λ is the forgetting factor which is set to 0.94 by the rule of thumb, and $\hat{\Sigma}_1 = \mathbf{0}$. Figure 3 shows the out-of-sample prediction errors in the Bures–Wasserstein and in the Frobenius distance. In both cases, RFM consistently yields competitive predictions, whereas LFM suffers from the largest errors. Both the LOCF and EWMA are slightly inferior to the RFM, indicating the proposed method is a viable alternative to these popular benchmarks.

Since covariance forecasting is often used in portfolio risk assessments, we also evaluate the methods in this context. For illustration, we consider the minimum variance portfolio $\mathbf{w}_t$ at each time t , which is given by $\mathbf{w}_t = \Sigma_t^{-1} \mathbf{1} / (\mathbf{1}^\top \Sigma_t^{-1} \mathbf{1})$ if short positions are allowed.

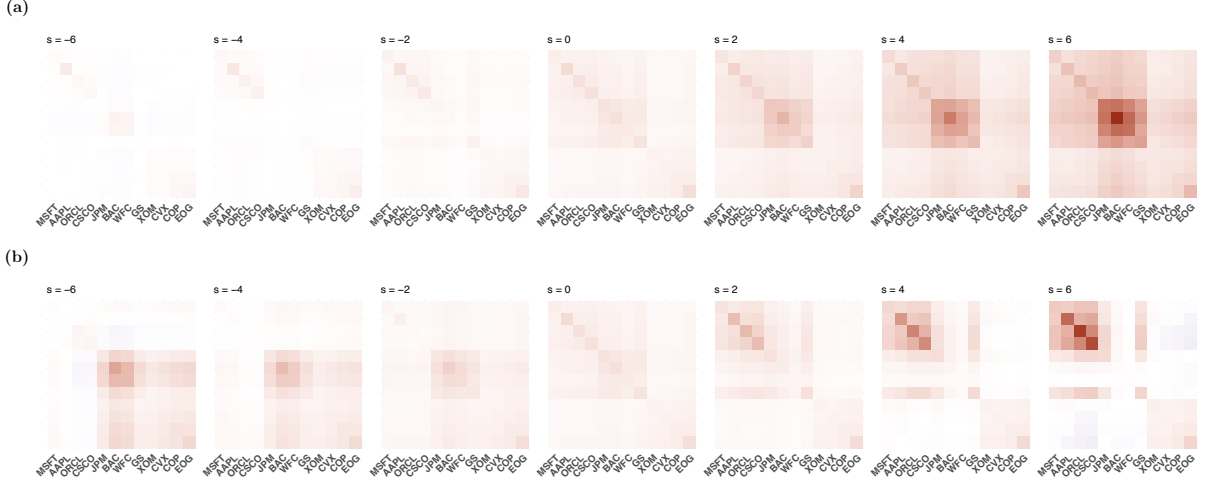


Figure 2: The loadings corresponding to the first factor (row (a)) and the second factor (row (b)) estimated from the U.S. return data, represented as the geodesic trajectory $\text{Exp}_{\hat{\Sigma}}(s\hat{\mathbf{a}}_j)$ for $s \in [-6, 6]$, $j = 1, 2$, on the Bures–Wasserstein manifold. Red and blue entries signify positive and negative values, with darker colors representing larger magnitude. The color white corresponds to zero.

The predicted risk of this portfolio for the following month is given by $\mathbf{w}_t^\top \hat{\Sigma}_{t+1} \mathbf{w}_t$. Figure 4 shows the risk prediction errors $|\mathbf{w}_t^\top (\hat{\Sigma}_{t+1} - \Sigma_{t+1}) \mathbf{w}_t|$ across the test period. The RFM consistently yields low prediction errors throughout, with a mean error of 0.94, substantially outperforming LFM (3.66), LOCF (2.61), and EWMA (1.45). The predictions of RFM are also quite robust, achieving a median error of 0.52, the lowest among all methods, compared to the second-best EWMA (0.89). These results demonstrate the effectiveness of RFM in risk assessments. To summarize, in the analysis of the realized volatility series, the RFM not only offers effective dimension reduction with interpretable factors, but also produces competitive out-of-sample forecasts. This, in turn, facilitates accurate risk assessments, corroborating the proposed method’s practical value.

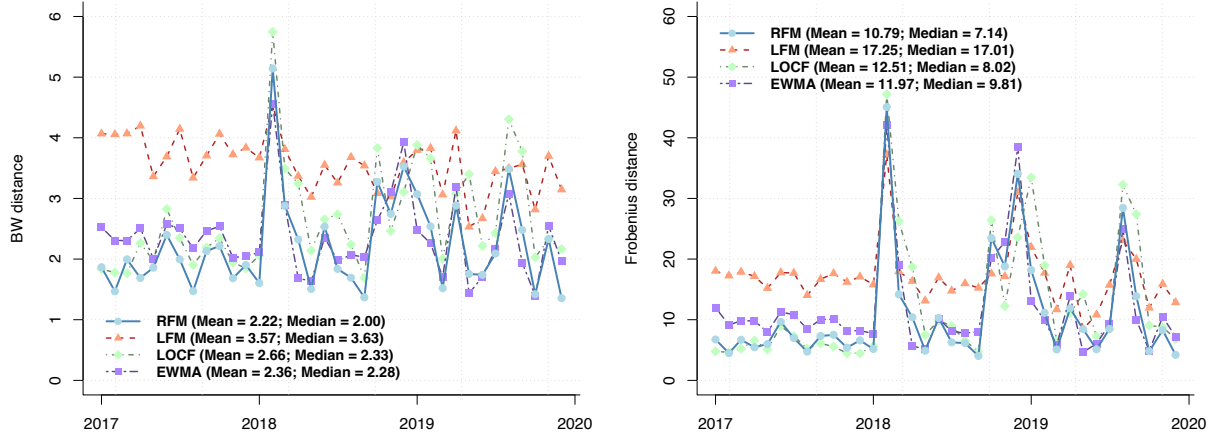


Figure 3: Out-of-sample prediction errors of RFM, LFM, LOCF, and EWMA. Left: In Bures–Wasserstein distance. Right: In Frobenius distance.

6 Conclusion

In this paper, we proposed the Riemannian factor model, a new factor modeling approach for time series in nonlinear manifolds. We established the convergence rates for the estimated loading space, which, under short-memory and strong factor conditions, can be as fast as the dimension-free rate $n^{-1/2}$, matching the ideal convergence rates for the conventional linear factor models. Numerical experiments, including simulation studies and an analysis with the realized covariance matrices of the U.S. stock returns, show that the RFM offers efficient dimension reduction, interpretable factors, and enhanced prediction capacity, which supports the RFM as a viable factor modeling approach for manifold-valued time series.

7 Disclosure statement

The authors report there are no competing interests to declare.

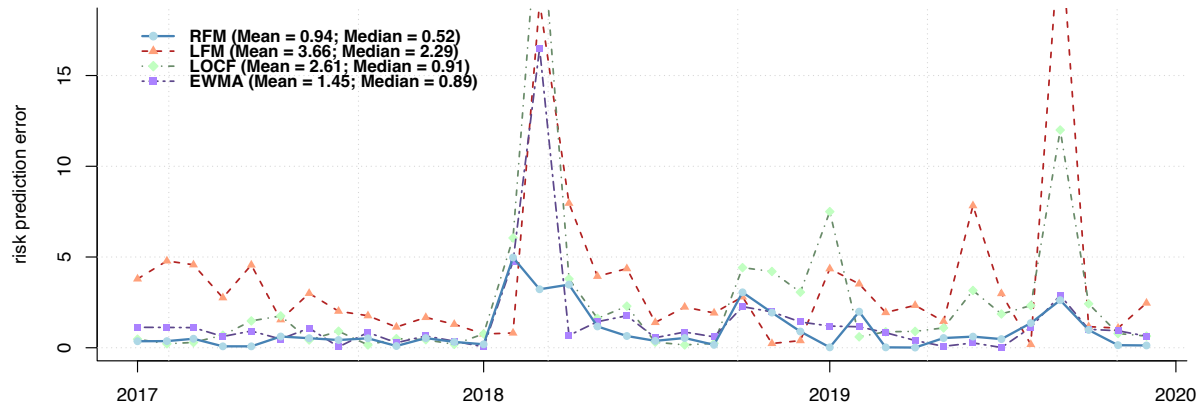


Figure 4: Out-of-sample risk prediction errors of RFM, LFM, LOCF, and EWMA for the minimum variance portfolios.

8 Data Availability Statement

The data used in the analysis were obtained from Yahoo! Finance, which cannot be released according to its API's terms of use. However, the codes for data acquisition and analysis are available in the Github repository [Omitted for anonymity of the review].

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Supplementary material to “A Riemannian Factor Model for Manifold-Valued Time Series”

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July 7, 2026

The supplementary material contains three sections. Section [S.1](#) contains the theoretical derivations of the main results as well as auxiliary lemmas. Section [S.2](#) collects some facts about the Bures–Wasserstein manifold, which, in particular, provides a way to construct orthonormal basis for the tangent spaces. Section [S.3](#) presents the simulation results when the data are generated on the Bures–Wasserstein manifold.

S.1 Technical derivations

This section contains the proofs of the results presented in Section [2](#) and Section [3](#) as well as auxiliary theoretical results.

S.1.1 Proofs for Section 2

PROOF OF PROPOSITION 1. Suppose (2) holds with (F), and let $\mathbf{E} = (\mathbf{e}_1, \dots, \mathbf{e}_p)$ be an arbitrary orthonormal basis of $T_\mu M$. Then $\boldsymbol{\delta}_{t,\mathbf{E}}^\top = (\langle \boldsymbol{\delta}_t, \mathbf{e}_1 \rangle_\mu, \dots, \langle \boldsymbol{\delta}_t, \mathbf{e}_p \rangle_\mu) =: (\delta_{t,\mathbf{E},1}, \dots, \delta_{t,\mathbf{E},p})$.

By (F), for any $i, j \in [p]$ and positive integer h , we have

$$\mathbb{E}(\delta_{t,\mathbf{E},i} \delta_{t-h,\mathbf{E},j}) = \mathbb{E}(\langle (\boldsymbol{\delta}_t \otimes \boldsymbol{\delta}_{t-h}) \mathbf{e}_i, \mathbf{e}_j \rangle) = 0. \quad (\text{S.1})$$

Define the matrix $\mathbf{A}_\mathbf{E}$ via

$$\mathbf{A}_\mathbf{E} \mathbf{g} = \begin{pmatrix} \langle \mathcal{A}(\mathbf{g}), \mathbf{e}_1 \rangle_\mu \\ \vdots \\ \langle \mathcal{A}(\mathbf{g}), \mathbf{e}_p \rangle_\mu \end{pmatrix}, \quad \mathbf{g} \in \mathbb{R}^r.$$

It is easy to verify that $(\mathbf{z}_{t,\mathbf{E}}, \mathbf{A}_\mathbf{E}, \boldsymbol{\delta}_{t,\mathbf{E}})$ satisfies (3). Since $\mathcal{A}$ is isometric, for any $\mathbf{g} \in \mathbb{R}^r$ we have

$$\|\mathbf{g}\|^2 = \|\mathcal{A}(\mathbf{g})\|_\mu^2 = \sum_{j=1}^p \langle \mathcal{A}(\mathbf{g}), \mathbf{e}_j \rangle_\mu^2 = \|\mathbf{A}_\mathbf{E} \mathbf{g}\|^2.$$

This implies $\mathbf{A}_\mathbf{E}$ has orthonormal columns by standard arguments, and, when combined with (S.1), proves (F'). Conversely, suppose $\mathbf{z}_{t,\mathbf{E}}$ admits the factor model (3) when expressed in some basis $\mathbf{E}$ such that (F') holds. Define $\mathbf{z}_t = z_{t,\mathbf{E},1} \mathbf{e}_1 + \dots + z_{t,\mathbf{E},p} \mathbf{e}_p \in T_\mu M$ and similarly $\boldsymbol{\delta}_t = \delta_{t,\mathbf{E},1} \mathbf{e}_1 + \dots + \delta_{t,\mathbf{E},p} \mathbf{e}_p \in T_\mu M$, where $\mathbf{z}_{t,\mathbf{E}} = (z_{t,\mathbf{E},1}, \dots, z_{t,\mathbf{E},p})^\top$ and $\boldsymbol{\delta}_{t,\mathbf{E}} = (\delta_{t,\mathbf{E},1}, \dots, \delta_{t,\mathbf{E},p})^\top$. In addition, define

$$\mathcal{A}(\mathbf{g}) = \tilde{g}_1 \mathbf{e}_1 + \dots + \tilde{g}_p \mathbf{e}_p, \quad \mathbf{g} \in \mathbb{R}^r,$$

where $\mathbf{A}_{\mathbf{E}}\mathbf{g} = (\tilde{g}_1, \dots, \tilde{g}_p)^\top$. Clearly $\mathcal{A}$ is linear and $\mathbf{z}_t = \mathcal{A}(\mathbf{f}_t) + \delta_t$. Now for any $\mathbf{v}, \mathbf{w} \in T_\mu M$ and positive integer h , we have

$$\begin{aligned} \mathbb{E}\{\langle (\delta_t \otimes \delta_{t-h})\mathbf{v}, \mathbf{w} \rangle\} &= \mathbb{E}\{\langle \delta_t, \mathbf{v} \rangle_\mu \langle \delta_{t-h}, \mathbf{w} \rangle_\mu\} \\ &= \begin{pmatrix} \langle \mathbf{v}, \mathbf{e}_1 \rangle_\mu & \dots & \langle \mathbf{v}, \mathbf{e}_p \rangle_\mu \end{pmatrix} \mathbb{E}(\delta_{t,\mathbf{E}} \delta_{t-h,\mathbf{E}}^\top) \begin{pmatrix} \langle \mathbf{w}, \mathbf{e}_1 \rangle_\mu \\ \vdots \\ \langle \mathbf{w}, \mathbf{e}_p \rangle_\mu \end{pmatrix} \\ &= 0, \end{aligned}$$

where the last equality follows from (F'). Note that $\|\mathcal{A}(\mathbf{g})\|_\mu^2 = \sum_{j=1}^p \tilde{g}_j^2 = \|\mathbf{A}_{\mathbf{E}}\mathbf{g}\|^2 = \|\mathbf{g}\|^2$, where the last equality also follows from (F'). Hence $\mathcal{A}$ is isometric and (F) is proved. $\square$

Before proving the claims in Example 1, we introduce the definition of α -mixing coefficients and review some of the basic properties.

Definition S.1. Let $\mathcal{A}, \mathcal{B}$ be two sub- σ -field of $\mathcal{F}$. Define $\alpha(\mathcal{A}, \mathcal{B}) = \sup_{A \in \mathcal{A}, B \in \mathcal{B}} |\mathbb{P}(A \cap B) - \mathbb{P}(A)\mathbb{P}(B)|$.

Definition S.2. A vector-valued process $\{\mathbf{z}_t\}_{t \in \mathbb{Z}}$ is α -mixing if

$$\alpha_k = \sup_t \alpha(\sigma\{\mathbf{z}_s : s \leq t\}, \sigma\{\mathbf{z}_s : s \geq t+k\}) \rightarrow 0, \quad (\text{S.2})$$

where $\sigma\{\mathbf{z}_s : s \in S\}$ denotes the σ -field generated by $\{\mathbf{z}_s : s \in S\}$. The numbers $\{\alpha_k\}$ are called the α -mixing coefficients of $\{\mathbf{z}_t\}$.

The following remarks are standard results in the literature.

Remark S.1. If $\mathbf{z}_t$ is strictly stationary, then “ $\sup_t$ ” can be dropped in (S.2).

Remark S.2. Let f be a real-valued measurable function. If $\{\mathbf{z}_t\}$ is α -mixing with α -mixing coefficients α_k , then the process $\{f(\mathbf{z}_t)\}$ is α -mixing with coefficients $\alpha_{f,k} \leq \alpha_k$.

Now we can prove the claims in Example 1.

PROOF OF EXAMPLE 1. For $\mathbf{w} \in B_\mu(R)$, write

$$f_{\mathbf{w}}(\mathbf{z}_t) = d_M(\mathbf{w}, \text{Exp}_\mu(\mathbf{z}_t))^2 - \mathbb{E}(d_M(\mathbf{w}, \text{Exp}_\mu(\mathbf{z}_t))^2).$$

Clearly, $\mathbb{E}f_{\mathbf{w}}(\mathbf{z}_t) = 0$ and $|f_{\mathbf{w}}(\mathbf{z}_t)| < 8R^2$ for all t almost surely by (P1). Moreover, the α -mixing coefficient $\alpha_{k,f}$ of $\{f_{\mathbf{w}}(\mathbf{z}_t)\}$ is upper bounded by α_k for all k . By Theorem 1.3 of Bosq (1998),

$$\mathbb{P}\left(\left|\frac{1}{n}\sum_{t=1}^n f_{\mathbf{w}}(\mathbf{z}_t)\right| > \delta\right) \leq 4\exp\left(-\frac{\delta^2 q}{512R^2}\right) + 22\left(1 + \frac{32R^2}{\delta}\right)^{1/2} q\alpha_{\lfloor \frac{n}{2q} \rfloor} \quad (\text{S.3})$$

for all integer $q \in [1, n/4]$ and $\delta > 0$. Fix $\epsilon > 0$ and let $\{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_{N_\epsilon}\}$ be a minimal ϵ -cover of the set $\{\mathbf{v} \in T_\mu M : \|\mathbf{v}\| < R\}$, where N_ϵ is the covering number. That is, for every $\mathbf{v} \in T_\mu M$ with $\|\mathbf{v}\| < R$ there exists some $j_{\mathbf{v}} \in [N_\epsilon]$ with $\|\mathbf{v} - \mathbf{w}_{j_{\mathbf{v}}}\| < \epsilon$. Note that $|f_{\mathbf{w}}(\mathbf{z}) - f_{\mathbf{w}_{j_{\mathbf{v}}}}(\mathbf{z})| \leq 8Rd_M(\mathbf{v}, \mathbf{w}_{j_{\mathbf{v}}})$ for all $\mathbf{z} \in T_\mu M$ with $\|\mathbf{z}\| < R$. By Proposition 4 of Lin and Yao (2019), since the sectional curvature of M is bounded below by some $K_1 \in \mathbb{R}$, there exists a constant $C_{K_1, R}$, depending only K_1 and R , such that $d_M(\mathbf{v}, \mathbf{w}_{j_{\mathbf{v}}}) \leq \sqrt{C_{K_1, R}}\|\mathbf{v} - \mathbf{w}_{j_{\mathbf{v}}}\| \leq \sqrt{C_{K_1, R}}\epsilon$. This, combined with (S.3), yields

$$\begin{aligned} & \mathbb{P}\left(\sup_{\mathbf{w} \in B_\mu(R)} \left|n^{-1}\sum_{t=1}^n f_{\mathbf{w}}(\mathbf{z}_t)\right| > 9R\sqrt{C_{K_1, R}}\epsilon\right) \\ & \leq \left(1 + \frac{2R}{\epsilon}\right)^p \left\{4\exp\left(-\frac{qC_{K_1, R}\epsilon^2}{512}\right) + 22\left(1 + \frac{32R}{\sqrt{C_{K_1, R}}\epsilon}\right)^{1/2} q\alpha_{\lfloor \frac{n}{2q} \rfloor}\right\}, \end{aligned} \quad (\text{S.4})$$

where we used the well-known fact that the ϵ -covering number of a ball of radius R in $\mathbb{R}^p$ is bounded by $(1 + 2R/\epsilon)^p$. From (S.4), the desired results in Example 1 can be derived. Specifically, for the geometric decay case, plugging $\epsilon = n^{-(1-2\gamma)/2}$ and $q = \lceil n^{1-\gamma} \rceil$ in (S.4),

we have, for sufficiently large n ,

$$\begin{aligned}
& \mathbb{P} \left(\sup_{\mathbf{w} \in B_{\mu}(R)} \left| n^{-1} \sum_{t=1}^n f_{\mathbf{w}}(\mathbf{z}_t) \right| > 9R\sqrt{C_{K_1,R}} n^{-(1-2\gamma)/2} \right) \\
& \leq 22 \exp \left\{ p \log(4R) + \frac{1}{2}(1-2\gamma)p \log n - \frac{C_{K_1,R}}{512} n^{-\gamma} \right\} + \\
& \quad 22 \exp \left\{ o\left(\frac{n^\gamma}{\log n}\right) + \frac{1}{2}(1-2\gamma)p \log n - n^{1-\gamma} \log \frac{1}{\rho} \right\} \\
& \rightarrow 0.
\end{aligned}$$

For the algebraic decay case, a similar argument applies with $\epsilon = n^{-\eta/(3/2+p+2\zeta)} \log n$ and $q = \lceil n^{2\zeta/(3/2+p+2\zeta)} \rceil$. Details are omitted. $\square$

Proposition S.1 below gives an estimate on the magnitude of the α -mixing coefficients of vector-valued linear processes. Its proof follows from the proof of Theorem 2.1 of [Pham and Tran \(1985\)](#) with careful accounting on the constants that may depend on the dimension.

Proposition S.1. *Let $\{\mathbf{e}_t\}$ be a sequence of independent random vectors with density function g_t , and let $\mathbf{z}_t = \sum_{j=0}^{\infty} \mathbf{A}_j \mathbf{e}_{t-j}$ be a linear process with $\mathbf{A}_0 = \mathbf{I}$. Assume that*

$$(i) \quad \sum_{j=0}^{\infty} \|\mathbf{A}_j\|_{op} < \infty \text{ and } \sum_{j=0}^{\infty} \mathbf{A}_j z^j \neq 0 \text{ for all } z \text{ with } |z| \leq 1;$$

$$(ii) \quad \sum_{j=1}^{\infty} c_j^{\delta/(1+\delta)} < \infty \text{ for some } \delta > 0, \text{ where } c_j = \sum_{k \geq j} \|\mathbf{A}_k\|_{op}.$$

In addition, assume there exist constants K_1, K_2 such that

$$(iii) \quad \int |g_t(\mathbf{v} - \mathbf{u}) - g_t(\mathbf{v})| d\mathbf{v} \leq K_1 \|\mathbf{u}\|;$$

$$(iv) \quad \mathbb{E} \|\mathbf{e}_t\|^\delta < K_2 \text{ for all } t.$$

Then the α -mixing coefficient of $\{\mathbf{z}_t\}$ satisfies

$$\alpha_k \leq 8 \left(K_1 \left[\sum_{i=0}^{\infty} \|\mathbf{B}_i\|_{op} \right] + K_2 \right) \sum_{j=k}^{\infty} c_j^{\frac{\delta}{1+\delta}},$$

where $(\sum_{j=0}^{\infty} \mathbf{A}_j z^j)^{-1} = \sum_{j=0}^{\infty} \mathbf{B}_j z^j$.

Proposition S.1 can be used to bound the α -mixing coefficients for various time series. In the following example, we use the Gaussian VAR(1) process to illustrate how the α -mixing coefficients may depend on the dimension. The proof is based on elementary properties of the Gaussian density.

Example S.1 (Gaussian VAR(1)). *Suppose $\mathbf{e}_t$ is a sequence of p -dimensional independent Gaussian random vectors with mean zero and identity covariance. Consider the VAR(1) process $\mathbf{z}_t = \mathbf{A}\mathbf{z}_{t-1} + \mathbf{e}_t$ with $\|\mathbf{A}\|_{op} \leq \bar{\rho} < 1$ for some (dimension-independent) constant $\bar{\rho}$. Then the α -mixing coefficients of $\mathbf{z}_t$, $\{\alpha_k\}$, satisfy*

$$\alpha_k \leq 8 \frac{\bar{\rho} + 1}{\sqrt{1 - \bar{\rho}}(1 - \sqrt{\bar{\rho}})} (\mathbb{E}\|\mathbf{e}_t\|)(\bar{\rho})^{k/2}.$$

Thus, the α -mixing coefficients of a Gaussian VAR(1) process decay geometrically. However, since it is well-known that $\mathbb{E}\|\mathbf{e}_t\| \asymp \sqrt{p}$ as $p \rightarrow \infty$, the serial dependence becomes stronger in the high-dimensional regime.

Remark S.3. *Although in Example 1 we treat H in the case of geometric decay as fixed, the result is unaffected if $H = O(p^\zeta)$ for $\zeta \geq 0$ as in Example S.1. This is because H only has logarithmic effects in the geometric decay case. In the case of algebraic decay, however, the convergence rate c_n will be further affected if H scales polynomially with the dimension.*

S.1.2 Proofs for Section 3

First, we prove Proposition 2. The proof idea is similar to that in Dai and Müller (2018) and Bhattacharya and Patrangenaru (2003), with the necessary modifications for tracking the dependence on the dimension.

PROOF OF PROPOSITION 2. Let $0 < \delta < \eta$ and put

$$E = B_\mu \left(\frac{(2 + \delta)R}{2 + \eta} \right).$$

By (P1), $\hat{\boldsymbol{\mu}}, \mathbf{x}_t \in E$ almost surely. By (P3), we have

$$\sup_{\mathbf{w} \in E} \left| n^{-1} \sum_{t=1}^n [d_M(\mathbf{x}_t, \mathbf{w})^2 - \mathbb{E}(d_M(\mathbf{x}_t, \mathbf{w})^2)] \right| = O_p(c_n). \quad (\text{S.5})$$

Note that by (P2) and Eq. (S.5),

$$\begin{aligned} & \left| n^{-1} \sum_{t=1}^n [d_M(\mathbf{x}_t, \hat{\boldsymbol{\mu}})^2 - \mathbb{E}(d_M(\mathbf{x}_t, \boldsymbol{\mu})^2)] \right| \\ & \leq \max \left\{ n^{-1} \sum_{t=1}^n [d_M(\mathbf{x}_t, \boldsymbol{\mu})^2 - \mathbb{E}(d_M(\mathbf{x}_t, \boldsymbol{\mu})^2)], n^{-1} \sum_{t=1}^n \left[\int d_M(\mathbf{x}, \hat{\boldsymbol{\mu}})^2 Q_t(d\mathbf{x}) - d_M(\mathbf{x}_t, \hat{\boldsymbol{\mu}})^2 \right] \right\} \\ & \leq \sup_{\mathbf{w} \in E} \left| n^{-1} \sum_{t=1}^n [d_M(\mathbf{x}_t, \mathbf{w})^2 - \mathbb{E}(d_M(\mathbf{x}_t, \mathbf{w})^2)] \right| \\ & = O_p(c_n). \end{aligned} \quad (\text{S.6})$$

If $\mathbf{y} \notin E$, we have $d_M(\mathbf{x}_t, \mathbf{y}) \geq d_M(\mathbf{y}, \boldsymbol{\mu}) - d_M(\boldsymbol{\mu}, \mathbf{x}_t) \geq (1 + \delta)R/(2 + \eta)$. On the other hand, for $\epsilon < \delta R/(2 + \eta)$ and $\mathbf{v} \in T_{\boldsymbol{\mu}}M$ with $\|\mathbf{v}\| = 1$, we have $d_M(\text{Exp}_{\boldsymbol{\mu}}(\epsilon \mathbf{v}), \mathbf{x}_t) \leq \epsilon + R/(2 + \eta) < (1 + \delta)R/(2 + \eta)$. Hence, for $\epsilon < \delta R/(2 + \eta)$,

$$\begin{aligned} & \inf_{\mathbf{w} \in M: d_M(\mathbf{w}, \boldsymbol{\mu}) > \epsilon} n^{-1} \sum_{t=1}^n [d_M(\mathbf{x}_t, \mathbf{w})^2 - \mathbb{E}(d_M(\mathbf{x}_t, \boldsymbol{\mu})^2)] \\ & = \inf_{\mathbf{w} \in E: d_M(\mathbf{w}, \boldsymbol{\mu}) > \epsilon} n^{-1} \sum_{t=1}^n [d_M(\mathbf{x}_t, \mathbf{w})^2 - \mathbb{E}(d_M(\mathbf{x}_t, \mathbf{w})^2) + \mathbb{E}(d_M(\mathbf{x}_t, \mathbf{w})^2) - \mathbb{E}(d_M(\mathbf{x}_t, \boldsymbol{\mu})^2)] \\ & \geq a_\epsilon - O_p(c_n), \end{aligned} \quad (\text{S.7})$$

where $a_\epsilon = \inf_{n \in \mathbb{N}} \inf_{t \in [n]} \inf_{\mathbf{w}: d_M(\mathbf{w}, \boldsymbol{\mu}) > \epsilon} \{\mathbb{E}(d_M(\mathbf{w}, \mathbf{x}_t)^2) - \mathbb{E}(d_M(\boldsymbol{\mu}, \mathbf{x}_t)^2)\} > 0$ by (P4), and the last inequality is due to (S.5). Finally, put

$$\begin{aligned} A = & \left\{ \left| n^{-1} \sum_{t=1}^n [d_M(\mathbf{x}_t, \hat{\boldsymbol{\mu}})^2 - \mathbb{E}(d_M(\mathbf{x}_t, \boldsymbol{\mu})^2)] \right| \leq \frac{a_\epsilon}{3} \right\} \cap \\ & \left\{ \inf_{\mathbf{w} \in M: d_M(\mathbf{w}, \boldsymbol{\mu}) > \epsilon} n^{-1} \sum_{t=1}^n [d_M(\mathbf{x}_t, \mathbf{w})^2 - \mathbb{E}(d_M(\mathbf{x}_t, \boldsymbol{\mu})^2)] \geq \frac{2}{3} a_\epsilon \right\}. \end{aligned}$$

Then by (S.6) and (S.7), $\mathbb{P}(d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) > \epsilon) \leq 1 - \mathbb{P}(A) = o(1)$, and (8) follows.

Next, we prove (9). On the event $\{d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) < \min\{B, R\}\}$, (P3) and the definition of

$\hat{\boldsymbol{\mu}}$ imply

$$\begin{aligned}
& n^{-1} \sum_{t=1}^n \int d_M(\mathbf{x}, \hat{\boldsymbol{\mu}})^2 Q_t(d\mathbf{x}) - n^{-1} \sum_{t=1}^n \int d_M(\mathbf{x}, \boldsymbol{\mu})^2 Q_t(d\mathbf{x}) \\
&= g_n(\text{Log}_{\boldsymbol{\mu}}(\hat{\boldsymbol{\mu}})) - g_n(\mathbf{0}) \\
&\leq n^{-1} \sum_{t=1}^n \int d_M(\mathbf{x}, \hat{\boldsymbol{\mu}})^2 Q_t(d\mathbf{x}) - n^{-1} \sum_{t=1}^n d_M(\mathbf{x}_t, \hat{\boldsymbol{\mu}})^2 \\
&\quad + n^{-1} \sum_{t=1}^n d_M(\mathbf{x}_t, \boldsymbol{\mu})^2 - n^{-1} \sum_{t=1}^n \int d_M(\mathbf{x}, \boldsymbol{\mu})^2 Q_t(d\mathbf{x}) \\
&= O_p(c_n),
\end{aligned} \tag{S.8}$$

where (P1) ensures $\text{Log}_{\boldsymbol{\mu}}(\hat{\boldsymbol{\mu}})$ is well-defined in the first equality. Since $\mathbf{0}$ is the minimizer of g_n over $\{\mathbf{v} \in T_{\boldsymbol{\mu}}M : \|\mathbf{v}\| < R\}$ by (P2), we have

$$g_n(\mathbf{v}) - g_n(\mathbf{0}) \geq \lambda_n \|\mathbf{v}\|^2, \quad \text{for } \|\mathbf{v}\| < \min\{B, R\}, \tag{S.9}$$

where $\lambda_n^{-1} = \sup_{\|\mathbf{v}\| < B} \lambda_{\min}^{-1}(\partial^2 g_n(\mathbf{v})/\partial \mathbf{v}^2) = O(1)$ by (P5). Combining (S.8) and (S.9) yields

$$\begin{aligned}
d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu})^2 \mathbb{I}\{d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) < \min\{B, R\}\} &= \|\text{Log}_{\boldsymbol{\mu}}(\hat{\boldsymbol{\mu}})\|^2 \mathbb{I}\{d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) < \min\{B, R\}\} \\
&\leq O_p(\lambda_n^{-1} c_n) = O_p(c_n),
\end{aligned}$$

where $\mathbb{I}\{A\}$ is the indicator of an event A . Since $\lim_{n \rightarrow \infty} \mathbb{P}(d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) < \min\{B, R\}) = 1$ by (8), this proves (9). □

In order to prove Theorem 1, which bounds the angle between the estimated loading space (after parallel transport) and the true loading space, we need to first study how the estimated Fréchet mean contaminates the log-mapped data. Theorem S.5 below serves this purpose. Before proving Theorem S.5, we need some auxiliary results.

Lemma S.2 (do Carmo, 1992, Prop. 2.5, Ch. 10). *Let M and $\tilde{M}$ be two p -dimensional Riemannian manifolds and suppose that for all $\mathbf{p} \in M$, $\tilde{\mathbf{p}} \in \tilde{M}$, and all two-dimensional*

subspaces $\sigma \subset T_{\mathbf{p}}M$, $\tilde{\sigma} \subset T_{\tilde{\mathbf{p}}}\tilde{M}$, we have $K(\tilde{\mathbf{p}}, \tilde{\sigma}) \geq K(\mathbf{p}, \sigma)$. Fix $\mathbf{p} \in M$, $\tilde{\mathbf{p}} \in \tilde{M}$ and a linear isometry $i : T_{\mathbf{p}}M \mapsto T_{\tilde{\mathbf{p}}}\tilde{M}$. Let $r > 0$ be such that $\text{Exp}_{\mathbf{p}}|_{B_0(r)}$ is a diffeomorphism and $\text{Exp}_{\tilde{\mathbf{p}}}|_{\tilde{B}_0(r)}$ is nonsingular. For a differentiable curve $c : I \mapsto \text{Exp}_{\mathbf{p}}(B_0(r))$, $I \subset \mathbb{R}$, define $\tilde{c} : I \mapsto \text{Exp}_{\tilde{\mathbf{p}}}(\tilde{B}_0(r))$ by

$$\tilde{c} = \text{Exp}_{\tilde{\mathbf{p}}} \circ i \circ \text{Log}_{\mathbf{p}} \circ c.$$

Then $\ell(c) \geq \ell(\tilde{c})$, where $\ell(c)$ denotes the length of the curve c .

In the following, $(d\text{Exp}_{\xi})_{\mathbf{v}} : T_{\mathbf{v}}T_{\xi}M \rightarrow T_{\text{Exp}_{\xi}(\mathbf{v})}M$, where $\xi \in M$ and $\mathbf{v} \in T_{\xi}M$, denotes the differential of Exp_{ξ} at $\mathbf{v}$, which is a linear operator.

Lemma S.3. *Let $\xi \in M$ and $\mathbf{v} \in T_{\xi}M$ such that $\|\mathbf{v}\| < \rho < \bar{\rho}$, where $\bar{\rho}$ is the radius of injectivity at ξ . If the sectional curvature on $B_{\xi}(\rho)$ is bounded below by $K_1 > -\infty$, then*

$$\|(d\text{Exp}_{\xi})_{\mathbf{v}}\|_{op} \begin{cases} \leq 1 & \text{if } K_1 \geq 0 \\ \leq \frac{\sinh(\rho\sqrt{-K_1})}{\rho\sqrt{-K_1}} & \text{if } K_1 < 0 \end{cases}. \quad (\text{S.10})$$

On the other hand, if the sectional curvature on $B_{\xi}(\rho)$ is bounded above by a positive constant $K_0 < \infty$ and if $\rho < \pi/\sqrt{K_0}$, then

$$\|(d\text{Exp}_{\xi})_{\mathbf{v}}\mathbf{u}\| \geq \frac{\sin(\rho\sqrt{K_0})}{\rho\sqrt{K_0}}\|\mathbf{u}\|, \quad (\text{S.11})$$

for all $\mathbf{u} \in T_{\mathbf{v}}T_{\xi}M \simeq T_{\xi}M$.

Proof. In the following we prove (S.11), and (S.10) follows from a similar argument. Let $\mathbf{u} \in T_{\mathbf{v}}T_{\xi}M \simeq T_{\xi}M$ with $\|\mathbf{u}\| < \rho - \|\mathbf{v}\|$ and $J(t) = (d\text{Exp}_{\xi})_{t\mathbf{v}}(t\mathbf{u})$. It is easy to see that J is a Jacobi field along the geodesic $\gamma : t \mapsto \text{Exp}_{\xi}(t\mathbf{v})$, $t \in [0, 1]$. We can construct a Jacobi field on the model space $\tilde{M} = K_0^{-1/2}\mathbb{S}^p$, where $\mathbb{S}^p$ is the p -dimensional unit sphere, as follows. Let $\tilde{\mathbf{p}} \in \tilde{M}$ be arbitrary and let $\tilde{J}(t) = (d\text{Exp}_{\tilde{\mathbf{p}}})_{t\tilde{\mathbf{v}}}(t\tilde{\mathbf{u}})$, where $\tilde{\mathbf{v}} = i \circ \mathbf{v}$, $\tilde{\mathbf{u}} = i \circ \mathbf{u}$, and $i : T_{\xi}M \rightarrow T_{\tilde{\mathbf{p}}}\tilde{M}$ is an isometry. Clearly $\tilde{J}$ is a Jacobi field along the geodesic $\tilde{\gamma} : t \mapsto \text{Exp}_{\tilde{\mathbf{p}}}(t\tilde{\mathbf{v}})$, and $\tilde{\gamma}$ does not have conjugate points because $\|\tilde{\mathbf{v}}\| < \pi/\sqrt{K_0}$.

Since the sectional curvature along the geodesic γ is bounded above by that along $\tilde{\gamma}$ (which is constant K_0), it follows from Rauch comparison theorem (do Carmo, 1992, Thm. 2.3, Ch. 10) that

$$\|J(1)\| = \|(\mathrm{dExp}_\xi)_\mathbf{v} \mathbf{u}\| \geq \|\tilde{J}(1)\|. \quad (\text{S.12})$$

Decompose $\tilde{\mathbf{u}} = \tilde{\mathbf{u}}_T + \tilde{\mathbf{u}}_N$, where $\tilde{\mathbf{u}}_T = (\langle \tilde{\mathbf{u}}, \tilde{\mathbf{v}} \rangle / \|\tilde{\mathbf{v}}\|^2) \tilde{\mathbf{v}}$ and $\tilde{\mathbf{u}}_N = \tilde{\mathbf{u}} - \tilde{\mathbf{u}}_T$. Then

$$\begin{aligned} \tilde{J}(t) &= (\mathrm{dExp}_{\tilde{\mathbf{p}}})_{t\tilde{\mathbf{v}}}(t\tilde{\mathbf{u}}_T) + (\mathrm{dExp}_{\tilde{\mathbf{p}}})_{t\tilde{\mathbf{v}}}(t\tilde{\mathbf{u}}_N) \\ &=: \tilde{J}_T(t) + \tilde{J}_N(t). \end{aligned}$$

It is easy to verify that $\tilde{J}_T$ and $\tilde{J}_N$ are Jacobi fields along $\tilde{\gamma}$ and that $\tilde{J}_N$ is orthogonal to the velocity field of $\tilde{\gamma}$. Moreover, using the formula for normal Jacobi fields on manifolds with constant curvature, we have

$$\|\tilde{J}_N(1)\| = \left\| \frac{\sin(\|\tilde{\mathbf{v}}\|\sqrt{K_0})}{\|\tilde{\mathbf{v}}\|\sqrt{K_0}} \mathcal{P}_{\tilde{\mathbf{p}} \rightarrow \tilde{\gamma}(1)} \tilde{\mathbf{u}}_N \right\| \geq \frac{\sin(\rho\sqrt{K_0})}{\rho\sqrt{K_0}} \|\tilde{\mathbf{u}}_N\|. \quad (\text{S.13})$$

In addition, it follows from Gauss lemma (do Carmo, 1992, Lemma 3.5, Ch. 3) that

$$\|\tilde{J}_T(1)\| = \|\tilde{\mathbf{u}}_T\|. \quad (\text{S.14})$$

Combining (S.12)–(S.14) yields (S.11) for all $\mathbf{u}$ with $\|\mathbf{u}\| < \rho - \|\mathbf{v}\|$. Since $(\mathrm{dExp}_\xi)_\mathbf{v}$ is linear, the result extends to all $\mathbf{u} \in T_\xi M$. $\square$

Lemma S.4. *Let $\mathbf{x}, \mathbf{y}, \mathbf{z} \in M$ be such that any two of them lie within the radius of injectivity of the third point, and let $\gamma : [0, 1] \rightarrow M$ the geodesic joining $\mathbf{y}$ and $\mathbf{z}$. Assume*

- (i) *there exists some $0 < \rho < \bar{\rho}$ such that for all $\mathbf{w} \in B_\mathbf{x}(\rho)$ and all two-dimensional subspace $\sigma \subset T_\mathbf{w}M$,*

$$K_1 \leq K(\mathbf{w}, \sigma) \leq K_0,$$

where $\bar{\rho}$ is the radius of injectivity at $\mathbf{x}$;

(ii) for all $t \in [0, 1]$, $d_M(\mathbf{x}, \gamma(t)) < \rho < \pi/\sqrt{K_0}$.

Then there exists a constant $C < \infty$, depending only on K_0, K_1 , and ρ , such that

$$\text{Area}(\triangle \mathbf{x}\mathbf{y}\mathbf{z}) \leq C d_M(\mathbf{y}, \mathbf{z}),$$

where $\text{Area}(\triangle \mathbf{x}\mathbf{y}\mathbf{z})$ is the area of the geodesic triangle.

Proof. Without loss of generality, we may take $K_0 \geq 0$ and $K_1 \leq 0$. Since the geodesic triangle $\triangle = \triangle \mathbf{x}\mathbf{y}\mathbf{z}$ entirely lies in a normal neighborhood of $\mathbf{x}$ by (ii), we may parametrize $\triangle$ by

$$f(t, s) = \text{Exp}_{\mathbf{x}}(t \text{Log}_{\mathbf{x}} \gamma(s)),$$

where $\gamma(s) = \text{Exp}_{\mathbf{y}}(s \text{Log}_{\mathbf{y}} \mathbf{z})$ is the geodesic joining $\mathbf{y}$ and $\mathbf{z}$. By Lemma S.3, we have

$$\left\| \frac{\partial f}{\partial t} \right\| = \|(\text{dExp}_{\mathbf{x}})_{t \text{Log}_{\mathbf{x}} \gamma(s)}(\text{Log}_{\mathbf{x}} \gamma(s))\| \leq \frac{\sinh(\rho\sqrt{-K_1})}{\rho\sqrt{-K_1}} \|\text{Log}_{\mathbf{x}} \gamma(s)\| < \frac{\sinh(\rho\sqrt{-K_1})}{\rho\sqrt{-K_1}} \rho. \quad (\text{S.15})$$

and

$$\begin{aligned} \left\| \frac{\partial f}{\partial s} \right\| &= \left\| (\text{dExp}_{\mathbf{x}})_{t \text{Log}_{\mathbf{x}} \gamma(s)} \left(t (\text{dLog}_{\mathbf{x}})_{\gamma(s)} \frac{d\gamma(s)}{ds} \right) \right\| \\ &\leq t \frac{\sinh(\rho\sqrt{-K_1})}{\rho\sqrt{-K_1}} \left\| (\text{dExp}_{\mathbf{x}})_{\text{Log}_{\mathbf{x}} \gamma(s)}^{-1} \frac{d\gamma(s)}{ds} \right\| \\ &\leq t \frac{\sinh(\rho\sqrt{-K_1})}{\rho\sqrt{-K_1}} \frac{\rho\sqrt{K_0}}{\sin(\rho\sqrt{K_0})} \left\| \frac{d\gamma(s)}{ds} \right\|. \end{aligned} \quad (\text{S.16})$$

It follows from (S.15)–(S.16) that

$$\begin{aligned} \text{Area}(\triangle \mathbf{x}\mathbf{y}\mathbf{z}) &= \int_0^1 \int_0^1 \left\| \frac{\partial f}{\partial t} \wedge \frac{\partial f}{\partial s} \right\| ds dt \\ &\leq \frac{\rho}{2} \left(\frac{\sinh(\rho\sqrt{-K_1})}{\rho\sqrt{-K_1}} \right)^2 \frac{\rho\sqrt{K_0}}{\sin(\rho\sqrt{K_0})} \int_0^1 \left\| \frac{d\gamma(s)}{ds} \right\| ds \\ &= \frac{\rho}{2} \left(\frac{\sinh(\rho\sqrt{-K_1})}{\rho\sqrt{-K_1}} \right)^2 \frac{\rho\sqrt{K_0}}{\sin(\rho\sqrt{K_0})} d_M(\mathbf{y}, \mathbf{z}), \end{aligned}$$

where $\|\mathbf{u} \wedge \mathbf{v}\| = \sqrt{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2 - \langle \mathbf{u}, \mathbf{v} \rangle^2}$, which completes the proof. $\square$

Theorem S.5. Assume (M1), (P1)–(P5), and that there exists some $0 < \delta < \eta$ such that the radius of injectivity of every point of $B_{\boldsymbol{\mu}}(R/(2 + \eta))$ is at least $(1 + \delta)R/(2 + \eta)$. If $R < \left(\frac{2+\eta}{1+\delta}\right)\frac{\pi}{\sqrt{K_0}}$, then the log-mapped data $\hat{\mathbf{z}}_t = \text{Log}_{\hat{\boldsymbol{\mu}}}(\mathbf{x}_t)$ satisfy

$$\max_{t=1,2,\dots,n} \|\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}} \hat{\mathbf{z}}_t - \mathbf{z}_t\|_{\boldsymbol{\mu}} = O_p(c_n^{1/2}). \quad (\text{S.17})$$

Moreover, for all $h = 1, 2, \dots, h_0$,

$$\|\Phi_{\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}} \hat{\mathcal{S}}(h) - \mathcal{S}(h)\|_{op} = O_p(c_n^{1/2}), \quad \|\Phi_{\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}} \hat{\mathcal{S}}(h)^* - \mathcal{S}(h)^*\|_{op} = O_p(c_n^{1/2}), \quad (\text{S.18})$$

where h_0 is a fixed positive integer and $\mathcal{S}(h) = n^{-1} \sum_{t=h_0+1}^n \mathbf{z}_{t-h} \otimes \mathbf{z}_t$.

Proof. By Proposition 2, $\lim_{n \rightarrow \infty} \mathbb{P}(d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) < \delta R/(2 + \eta)) = 1$. Hence we consider the case $d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) < \delta R/(2 + \eta)$ in the rest of the proof. In particular, this implies $d_M(\mathbf{x}_t, \hat{\boldsymbol{\mu}}) < (1 + \delta)R/(2 + \eta)$ and therefore there exists a unique geodesic joining $\mathbf{x}_t$ and $\hat{\boldsymbol{\mu}}$ for all t . Since parallel transport is an isometry, we have

$$\begin{aligned} \|\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}} \text{Log}_{\hat{\boldsymbol{\mu}}} \mathbf{x}_t - \text{Log}_{\boldsymbol{\mu}} \mathbf{x}_t\|_{\boldsymbol{\mu}} &\leq \|\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}} \text{Log}_{\hat{\boldsymbol{\mu}}} \mathbf{x}_t - \mathcal{P}_{\mathbf{x}_t \rightarrow \boldsymbol{\mu}} \mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \mathbf{x}_t} \text{Log}_{\hat{\boldsymbol{\mu}}} \mathbf{x}_t\|_{\boldsymbol{\mu}} + \\ &\quad \|\mathcal{P}_{\mathbf{x}_t \rightarrow \boldsymbol{\mu}} \mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \mathbf{x}_t} \text{Log}_{\hat{\boldsymbol{\mu}}} \mathbf{x}_t - \text{Log}_{\boldsymbol{\mu}} \mathbf{x}_t\|_{\boldsymbol{\mu}} \\ &= \|\text{Log}_{\hat{\boldsymbol{\mu}}} \mathbf{x}_t - \mathcal{P}_{\boldsymbol{\mu} \rightarrow \hat{\boldsymbol{\mu}}} \mathcal{P}_{\mathbf{x}_t \rightarrow \boldsymbol{\mu}} \mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \mathbf{x}_t} \text{Log}_{\hat{\boldsymbol{\mu}}} \mathbf{x}_t\|_{\hat{\boldsymbol{\mu}}} + \\ &\quad \|\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \mathbf{x}_t} \text{Log}_{\hat{\boldsymbol{\mu}}} \mathbf{x}_t - \mathcal{P}_{\boldsymbol{\mu} \rightarrow \mathbf{x}_t} \text{Log}_{\boldsymbol{\mu}} \mathbf{x}_t\|_{\mathbf{x}_t} \\ &=: (I) + (II). \end{aligned} \quad (\text{S.19})$$

To bound (II), note that by the uniqueness of parallel vector fields,

$$\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \mathbf{x}_t} \text{Log}_{\hat{\boldsymbol{\mu}}}(\mathbf{x}_t) = -\text{Log}_{\mathbf{x}_t}(\hat{\boldsymbol{\mu}}), \quad \mathcal{P}_{\boldsymbol{\mu} \rightarrow \mathbf{x}_t} \text{Log}_{\boldsymbol{\mu}}(\mathbf{x}_t) = -\text{Log}_{\mathbf{x}_t}(\boldsymbol{\mu}).$$

Hence $(II) = \|\text{Log}_{\mathbf{x}_t}(\hat{\boldsymbol{\mu}}) - \text{Log}_{\mathbf{x}_t}(\boldsymbol{\mu})\|_{\mathbf{x}_t}$. Then we compare M with a scaled sphere by applying Lemma S.2. Let $\tilde{M} = K_0^{-1/2} \mathbb{S}^p$ and $i : T_{\mathbf{x}_t} M \mapsto T_{\tilde{\mathbf{p}}} \tilde{M}$ be an isometry, where $\mathbb{S}^p$ is the p -dimensional unit sphere and $\tilde{\mathbf{p}} \in \tilde{M}$ is arbitrary. Let c be the geodesic on M with $c(0) = \boldsymbol{\mu}$, $c(1) = \hat{\boldsymbol{\mu}}$, and α the geodesic joining $\text{Exp}_{\tilde{\mathbf{p}}} \circ i \circ \text{Log}_{\mathbf{x}_t}(\boldsymbol{\mu})$ and $\text{Exp}_{\tilde{\mathbf{p}}} \circ i \circ \text{Log}_{\mathbf{x}_t}(\hat{\boldsymbol{\mu}})$ on

$\tilde{M}$. Observe that since from the assumptions we have $\boldsymbol{\mu}, \hat{\boldsymbol{\mu}} \in B_{\mathbf{x}_t}((1+\delta)R/(2+\eta))$ and $(1+\delta)R/(2+\eta) < \pi/\sqrt{K_0}$, $\text{Exp}_{\mathbf{x}_t}|_{B_0((1+\delta)R/(2+\eta))}$ and $\text{Exp}_{\tilde{\mathbf{p}}}|_{\tilde{B}_0((1+\delta)R/(2+\eta))}$ are diffeomorphisms. Then Lemma S.2 implies $d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) \geq \ell(\alpha) =: \tilde{\ell}$. Let $\boldsymbol{\alpha}_0 = \text{Exp}_{\tilde{\mathbf{p}}} \circ i \circ \text{Log}_{\mathbf{x}_t}(\boldsymbol{\mu})$, $\boldsymbol{\alpha}_1 = \text{Exp}_{\tilde{\mathbf{p}}} \circ i \circ \text{Log}_{\mathbf{x}_t}(\hat{\boldsymbol{\mu}})$. Since i is an isometry, we have $d_{\tilde{M}}(\tilde{\mathbf{p}}, \boldsymbol{\alpha}_0) = \|\text{Log}_{\mathbf{x}_t}(\boldsymbol{\mu})\|_{\mathbf{x}_t} =: a$. Similarly, $d_{\tilde{M}}(\tilde{\mathbf{p}}, \boldsymbol{\alpha}_1) = \|\text{Log}_{\mathbf{x}_t}(\hat{\boldsymbol{\mu}})\|_{\mathbf{x}_t} =: b$. By the spherical law of cosine (Zwillinger, 2018, Section 4.10.2.2), we have

$$\cos(\sqrt{K_0}\tilde{\ell}) = \cos(\sqrt{K_0}a)\cos(\sqrt{K_0}b) + \sin(\sqrt{K_0}a)\sin(\sqrt{K_0}b)\cos\theta,$$

where θ is the angle between $\text{Log}_{\tilde{\mathbf{p}}}(\boldsymbol{\alpha}_0)$ and $\text{Log}_{\tilde{\mathbf{p}}}(\boldsymbol{\alpha}_1)$. After some algebraic manipulations with the trigonometric identities, we have

$$\sin(\sqrt{K_0}a)\sin(\sqrt{K_0}b)\sin^2\frac{\theta}{2} = \sin\left(\frac{\sqrt{K_0}(\tilde{\ell}+b-a)}{2}\right)\sin\left(\frac{\sqrt{K_0}(\tilde{\ell}+a-b)}{2}\right). \quad (\text{S.20})$$

On the other hand, by the law of cosine for inner product spaces,

$$\begin{aligned} \|\text{Log}_{\mathbf{x}_t}(\hat{\boldsymbol{\mu}}) - \text{Log}_{\mathbf{x}_t}(\boldsymbol{\mu})\|_{\mathbf{x}_t}^2 &= \|\text{Log}_{\tilde{\mathbf{p}}}(\boldsymbol{\alpha}_1) - \text{Log}_{\tilde{\mathbf{p}}}(\boldsymbol{\alpha}_0)\|_{\tilde{\mathbf{p}}}^2 \\ &= a^2 + b^2 - 2ab\cos\theta \\ &= (a-b)^2 + 4ab\sin^2\frac{\theta}{2}. \end{aligned} \quad (\text{S.21})$$

Combining (S.20) and (S.21) yields

$$\begin{aligned} \|\text{Log}_{\mathbf{x}_t}(\hat{\boldsymbol{\mu}}) - \text{Log}_{\mathbf{x}_t}(\boldsymbol{\mu})\|_{\mathbf{x}_t}^2 &= (a-b)^2 + 4ab \frac{\sin\left(\frac{\sqrt{K_0}}{2}(\tilde{\ell}+b-a)\right)\sin\left(\frac{\sqrt{K_0}}{2}(\tilde{\ell}+a-b)\right)}{\sin(\sqrt{K_0}a)\sin(\sqrt{K_0}b)} \\ &\leq (a-b)^2 + \frac{(\sqrt{K_0}a)(\sqrt{K_0}b)}{\sin(\sqrt{K_0}a)\sin(\sqrt{K_0}b)}(\tilde{\ell}+b-a)(\tilde{\ell}+a-b) \\ &\leq \left(1 + \frac{4(\sqrt{K_0}a)(\sqrt{K_0}b)}{\sin(\sqrt{K_0}a)\sin(\sqrt{K_0}b)}\right) d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu})^2, \end{aligned}$$

where the first inequality follows from the facts that $\sin x \leq x$ for $x \geq 0$, $\min\{\tilde{\ell}-b+a, \tilde{\ell}-a+b\} \geq 0$ by the triangle inequality, and $\max\{\sqrt{K_0}a, \sqrt{K_0}b\} < \pi$. The second inequality above follows from $|a-b| = |d_M(\mathbf{x}_t, \boldsymbol{\mu}) - d_M(\mathbf{x}_t, \hat{\boldsymbol{\mu}})| \leq d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu})$ and $|\tilde{\ell} - (b-a)| \leq 2d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu})$.

Since $a \leq R/(2 + \eta)$ and $b \leq (1 + \delta)R/(2 + \eta)$, the above inequality implies

$$(II) \leq \left(1 + \frac{4(\sqrt{K_0}R/(2 + \eta))((1 + \delta)\sqrt{K_0}R/(2 + \eta))}{\sin(\sqrt{K_0}R/(2 + \eta))\sin((1 + \delta)\sqrt{K_0}R/(2 + \eta))}\right)^{1/2} d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}). \quad (\text{S.22})$$

For (I), note that $\text{Log}_{\hat{\boldsymbol{\mu}}}\mathbf{x}_t - \mathcal{P}_{\boldsymbol{\mu} \rightarrow \hat{\boldsymbol{\mu}}}\mathcal{P}_{\mathbf{x}_t \rightarrow \boldsymbol{\mu}}\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \mathbf{x}_t}\text{Log}_{\hat{\boldsymbol{\mu}}}\mathbf{x}_t$ is the difference between the tangent vector $\text{Log}_{\hat{\boldsymbol{\mu}}}\mathbf{x}_t$ and the tangent vector obtained by parallel transporting $\text{Log}_{\hat{\boldsymbol{\mu}}}\mathbf{x}_t$ along the loop formed by the geodesic triangle Δ with three vertices $\boldsymbol{\mu}$, $\hat{\boldsymbol{\mu}}$ and $\mathbf{x}_t$. By Lemma S.4, we have $\text{Area}(\Delta) \leq C_1 d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu})$, where C_1 depends only on K_0 , K_1 , and $\rho < (1 + \delta)R/(2 + \eta)$. Since the sectional curvature on M is bounded above in $B_{\boldsymbol{\mu}}(R)$, it follows from 6.1.1 and 6.2.1 of Buser and Karcher (1981) that

$$(I) \leq \|\text{Log}_{\hat{\boldsymbol{\mu}}}\mathbf{x}_t\| \frac{4}{3} \max\{|K_0|, |K_1|\} \text{Area}(\Delta) \leq \frac{4}{3} \frac{(1 + \delta)R}{2 + \eta} \max\{|K_0|, |K_1|\} \text{Area}(\Delta).$$

Hence we conclude that for some C_3 , depending on R , K_0 , K_1 , but not on the dimension p ,

$$(I) \leq C_3 d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}). \quad (\text{S.23})$$

Combining (S.19), (S.22), (S.23), and Proposition 2 yields (S.17).

For (S.18), note that $\Phi_{\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}}\hat{\mathcal{S}}(h) = n^{-1} \sum_{t=h_0+1}^n (\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}}\hat{\mathbf{z}}_{t-h}) \otimes (\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}}\hat{\mathbf{z}}_t)$. Observe that

$$\begin{aligned} \|\Phi_{\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}}\hat{\mathcal{S}}(h) - \mathcal{S}(h)\|_{op} &\leq \left\| n^{-1} \sum_{t=h_0+1}^n (\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}}\hat{\mathbf{z}}_{t-h} - \mathbf{z}_{t-h}) \otimes \mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}}\hat{\mathbf{z}}_t \right\|_{op} \\ &\quad + \left\| n^{-1} \sum_{t=h_0+1}^n \mathbf{z}_{t-h} \otimes (\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}}\hat{\mathbf{z}}_t - \mathbf{z}_t) \right\|_{op}. \end{aligned} \quad (\text{S.24})$$

Since $\max\{\|\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}}\hat{\mathbf{z}}_t\|, \|\mathbf{z}_t\| : t = 1, 2, \dots, n\} \leq (1 + \delta)R/(2 + \eta)$ almost surely, the first part of (S.18) follows from (S.24) and (S.17). Finally, since $\hat{\mathcal{S}}(h)^* = n^{-1} \sum_{t=h_0+1}^n \hat{\mathbf{z}}_t \otimes \hat{\mathbf{z}}_{t-h}$, the second part of (S.18) follows similarly. $\square$

Equipped with Theorem S.5, we are in the position to prove Theorem 1.

PROOF OF THEOREM 1. Note that $\Phi_{\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}}\hat{\mathcal{L}} = \sum_{h=1}^{h_0} (\Phi_{\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}}\hat{\mathcal{S}}(h))(\Phi_{\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}}\hat{\mathcal{S}}(h)^*)$. This, together with (S.18) and a similar telescoping argument used in Theorem S.5, proves (10).

For (12), we first analyze the eigengap of $\mathbf{\Lambda} = \sum_{h=1}^{h_0} \mathbb{E}(\mathbf{z}_{t-h} \otimes \mathbf{z}_t) \mathbb{E}(\mathbf{z}_{t-h} \otimes \mathbf{z}_t)^*$. We may assume $\kappa > 0$ because otherwise the upper bound is trivial. Since $\mathbf{z}_t = \mathcal{A}(\mathbf{f}_t) + \boldsymbol{\delta}_t$, we have

$$\mathbf{\Lambda} = \mathcal{A} \left\{ \sum_{h=1}^{h_0} (\boldsymbol{\Sigma}_f(h) \mathcal{A}^*) (\boldsymbol{\Sigma}_f(h) \mathcal{A}^*)^* \right\} \mathcal{A}^* := \mathcal{A} \mathcal{M} \mathcal{A}^*,$$

where $\boldsymbol{\Sigma}_f(h) = \mathbb{E}(\mathbf{f}_{t-h} \otimes \mathbf{f}_t)$. Clearly, $\mathbf{\Lambda}$ is nonnegative definite and self-adjoint, whose eigenspace is a subspace of $\text{Im}(\mathcal{A})$. Since $\mathcal{A}$ is an isometry,

$$\lambda_{\min}(\mathcal{M}) \geq \sum_{h=1}^{h_0} \lambda_{\min}(\boldsymbol{\Sigma}_f(h) \boldsymbol{\Sigma}_f(h)^*) \geq \kappa^2 > 0.$$

Hence, $\text{Im}(\mathbf{\Lambda}) = \text{Im}(\mathcal{A})$, and

$$\lambda_r(\mathbf{\Lambda}) \geq \kappa^2, \quad (\text{S.25})$$

where $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$ are the eigenvalues of $\mathbf{\Lambda}$. Next, we have

$$\begin{aligned} \|\mathcal{L} - \mathbf{\Lambda}\|_{op} &\leq \sum_{h=1}^{h_0} \left\| \frac{1}{n} \sum_{t=h_0+1}^n \mathbf{z}_{t-h} \otimes \mathbf{z}_t - \mathbb{E}(\mathbf{z}_{t-h} \otimes \mathbf{z}_t) \right\|_{op} \|\mathcal{S}(h)\|_{op} \\ &\quad + \sum_{h=1}^{h_0} \left\| \frac{1}{n} \sum_{t=h_0+1}^n \mathbf{z}_t \otimes \mathbf{z}_{t-h} - \mathbb{E}(\mathbf{z}_t \otimes \mathbf{z}_{t-h}) \right\|_{op} \|\mathbb{E}(\mathbf{z}_{t-h} \otimes \mathbf{z}_t)\|_{op}. \end{aligned}$$

Using the Hilbert-Schmidt norm to upper-bound the operator norm, it follows from (11) that $\|n^{-1} \sum_{t=h_0+1}^n \mathbf{z}_{t-h} \otimes \mathbf{z}_t - \mathbb{E}(\mathbf{z}_{t-h} \otimes \mathbf{z}_t)\|_{op} = O_p(\epsilon_n)$. Since $\|\mathbf{z}_t\| \leq R/(2 + \eta)$ a.s. by (P1), it follows that

$$\|\mathcal{L} - \mathbf{\Lambda}\|_{op} = O_p(\epsilon_n). \quad (\text{S.26})$$

Note that if $\hat{\mathbf{a}}_j$ is an eigenvector of $\hat{\mathcal{L}}$ associated with the eigenvalue $\hat{\lambda}_j$, then $\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}} \hat{\mathbf{a}}_j$ is an eigenvector of $\Phi_{\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}} \hat{\mathcal{L}}$ with the same eigenvalue. By (10) and (S.26), $\|\Phi_{\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}} \hat{\mathcal{L}} - \mathbf{\Lambda}\|_{op} = O_p(\sqrt{c_n} + \epsilon_n)$. This, combined with (S.25) and the Davis-Kahan theorem (Davis and Kahan, 1970), yields (12). $\square$

PROOF OF THEOREM 2. We first show

$$\begin{aligned} & \left\| \frac{1}{n} \sum_{t=h+1}^n \mathbf{z}_t \otimes \mathbf{z}_{t-h} - \mathbb{E}(\mathbf{z}_t \otimes \mathbf{z}_{t-h}) \right\|_{\text{HS}}^2 \\ &= \frac{(n-h)^2}{n^2} \sum_{i=1}^p \sum_{j=1}^p \left(\frac{1}{n-h} \sum_{t=h+1}^n (z_{t-h,i} z_{t,j} - \mathbb{E}(z_{t-h,i} z_{t,j})) \right)^2 = O_p(n^{-1}). \end{aligned} \quad (\text{S.27})$$

In the following, let C be a constant that does not depend on n (and hence p), whose value may vary in different lines. By Theorem A.2 of [Huang et al. \(2025\)](#) (see also the First Moment Bound Theorem of [Findley and Wei, 1993](#)), we have

$$\begin{aligned} & \sum_{i,j=1}^p \mathbb{E} \left\{ \frac{1}{n-h} \sum_{t=h+1}^n (z_{t-h,i} z_{t,j} - \mathbb{E}(z_{t-h,i} z_{t,j})) \right\}^2 \\ & \leq C \sum_{i,j=1}^p \frac{1}{(n-h)^2} \sum_{t=h+1}^n \sum_{s=h+1}^n \mathbb{E}(z_{t-h,i} z_{s-h,i}) \mathbb{E}(z_{t,j} z_{s,j}) \\ & = C \frac{1}{(n-h)^2} \sum_{t=h+1}^n \sum_{s=h+1}^n \text{Tr}(\mathbf{\Gamma}_{\boldsymbol{\mu}}(s-h, t-h)) \text{Tr}(\mathbf{\Gamma}_{\boldsymbol{\mu}}(s, t)) \\ & \leq C \frac{C_{\boldsymbol{\mu}}}{n-h} \left(\frac{R}{2+\eta} \right)^4 \sum_{k=-n-h+1}^{n-h-1} |k|^{-2d_{\boldsymbol{\mu}}} \\ & = O(n^{-1}), \end{aligned} \quad (\text{S.28})$$

where the second inequality follows from (13), (14), and that $\text{Tr}(\mathbf{\Gamma}(t, t)) = \sum_{i=1}^p \mathbb{E}(z_{t,i}^2) = \mathbb{E}(d_M(\mathbf{x}_t, \boldsymbol{\mu})^2) \leq (R/(2+\eta))^2$. Now (S.27) follows from Markov inequality and (S.28). This implies $\epsilon_n = n^{-1/2}$ in (11). Thus, it follows from the proof of Theorem S.5 and Theorem 1 (see Eq. (S.22) and Eq. (S.23)) that for (15) it suffices to show

$$d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) = O_p(n^{-1/2}). \quad (\text{S.29})$$

To show (S.29), recall from the proof of Proposition 2 that if $\|\hat{\mathbf{v}}\| < \min\{B, R\}$,

$$\begin{aligned} \lambda_n \|\hat{\mathbf{v}}\|^2 &\leq n^{-1} \sum_{t=1}^n \int d_M(\mathbf{x}, \hat{\boldsymbol{\mu}})^2 Q_t(d\mathbf{x}) - n^{-1} \sum_{t=1}^n d_M(\mathbf{x}_t, \hat{\boldsymbol{\mu}})^2 \\ &\quad + n^{-1} \sum_{t=1}^n d_M(\mathbf{x}_t, \hat{\boldsymbol{\mu}})^2 - n^{-1} \sum_{t=1}^n d_M(\mathbf{x}_t, \boldsymbol{\mu})^2 \\ &\quad + n^{-1} \sum_{t=1}^n d_M(\mathbf{x}_t, \boldsymbol{\mu})^2 - n^{-1} \sum_{t=1}^n \int d_M(\mathbf{x}, \boldsymbol{\mu})^2 Q_t(d\mathbf{x}) \\ &\leq \langle \nabla(g_n - f_n)(\tilde{\mathbf{v}}), \hat{\mathbf{v}} \rangle, \end{aligned}$$

where $\hat{\mathbf{v}} = \text{Log}_{\boldsymbol{\mu}}(\hat{\boldsymbol{\mu}})$, $\tilde{\mathbf{v}}$ lies on the line joining $\hat{\mathbf{v}}$ and $\mathbf{0}$ in $T_{\boldsymbol{\mu}}M$, and

$$f_n(\mathbf{v}) = n^{-1} \sum_{t=1}^n d_M(\mathbf{x}_t, \text{Exp}_{\boldsymbol{\mu}}(\mathbf{v}))^2, \quad g_n(\mathbf{v}) = n^{-1} \sum_{t=1}^n \int d_M(\mathbf{x}, \text{Exp}_{\boldsymbol{\mu}}(\mathbf{v})) Q_t(d\mathbf{x}).$$

Since $d_M(\hat{\boldsymbol{\mu}}, \boldsymbol{\mu}) = o_p(1)$, we may assume $\|\hat{\mathbf{v}}\| < \epsilon' = \min\{\epsilon, B, \delta R/(2 + \eta)\}$. Then by (P5)

it is sufficient to show

$$\sup_{\|\mathbf{v}\| < \epsilon'} \|\nabla(g_n - f_n)(\mathbf{v})\| = O_p(n^{-1/2}). \quad (\text{S.30})$$

Note that for $\mathbf{v}$ with $\|\mathbf{v}\| < \epsilon'$ and $\mathbf{x} \in B_{\boldsymbol{\mu}}(R/(2 + \eta))$, we have

$$\nabla d_M(\mathbf{x}, \text{Exp}_{\boldsymbol{\mu}}(\mathbf{v}))^2 = -2 \begin{pmatrix} \langle (\text{dExp}_{\boldsymbol{\mu}})_{\mathbf{v}} \mathbf{e}_1, \text{Log}_{\boldsymbol{\xi}}(\mathbf{x}) \rangle_{\boldsymbol{\xi}} \\ \langle (\text{dExp}_{\boldsymbol{\mu}})_{\mathbf{v}} \mathbf{e}_2, \text{Log}_{\boldsymbol{\xi}}(\mathbf{x}) \rangle_{\boldsymbol{\xi}} \\ \vdots \\ \langle (\text{dExp}_{\boldsymbol{\mu}})_{\mathbf{v}} \mathbf{e}_p, \text{Log}_{\boldsymbol{\xi}}(\mathbf{x}) \rangle_{\boldsymbol{\xi}} \end{pmatrix},$$

where $\boldsymbol{\xi} = \text{Exp}_{\boldsymbol{\mu}}(\mathbf{v})$. Differentiating under the integral yields

$$\begin{aligned} \|\nabla(g_n - f_n)(\mathbf{v})\|_{\boldsymbol{\mu}}^2 &= \left\| \frac{1}{n} \sum_{t=1}^n \int \nabla d_M(\mathbf{x}, \text{Exp}_{\boldsymbol{\mu}}(\mathbf{v}))^2 Q_t(d\mathbf{x}) - \frac{1}{n} \sum_{t=1}^n \nabla d_M(\mathbf{x}_t, \text{Exp}_{\boldsymbol{\mu}}(\mathbf{v}))^2 \right\|_{\boldsymbol{\mu}}^2 \\ &= 4 \sum_{j=1}^p \left\langle (\text{dExp}_{\boldsymbol{\mu}})_{\mathbf{v}} \mathbf{e}_j, n^{-1} \sum_{t=1}^n \left(\text{Log}_{\boldsymbol{\xi}}(\mathbf{x}_t) - \int \text{Log}_{\boldsymbol{\xi}}(\mathbf{x}) Q_t(d\mathbf{x}) \right) \right\rangle_{\boldsymbol{\xi}}^2 \\ &\leq 4 \|(\text{dExp}_{\boldsymbol{\mu}})_{\mathbf{v}}\|_{op}^2 \left\| n^{-1} \sum_{t=1}^n (\mathbf{z}_{t, \boldsymbol{\xi}} - \mathbb{E} \mathbf{z}_{t, \boldsymbol{\xi}}) \right\|_{\boldsymbol{\xi}}^2, \end{aligned} \quad (\text{S.31})$$

where the inequality follows since for any linear operator $\mathcal{T} : T_{\boldsymbol{\mu}}M \rightarrow T_{\boldsymbol{\xi}}M$ and vector $\mathbf{u} \in T_{\boldsymbol{\xi}}M$, $\sum_{j=1}^p \langle \mathcal{T} \mathbf{e}_j, \mathbf{u} \rangle^2 = \|\mathcal{T}^* \mathbf{u}\|_{\boldsymbol{\mu}}^2 \leq \|\mathcal{T}\|_{op}^2 \|\mathbf{u}\|_{\boldsymbol{\xi}}^2$.

Next, since $d_M(\boldsymbol{\xi}, \boldsymbol{\mu}) = \|\mathbf{v}\| < \epsilon'$ and

$$\mathbb{E} \left\| \sum_{t=1}^n (\mathbf{z}_{t,\boldsymbol{\xi}} - \mathbb{E} \mathbf{z}_{t,\boldsymbol{\xi}}) \right\|_{\boldsymbol{\xi}}^2 = \sum_{t=1}^n \sum_{s=1}^n \text{Tr}(\boldsymbol{\Gamma}_{\boldsymbol{\xi}}(s, t)),$$

it follows from Markov inequality and (13)–(14) that

$$\begin{aligned} \left\| n^{-1} \sum_{t=1}^n (\mathbf{z}_{t,\boldsymbol{\xi}} - \mathbb{E} \mathbf{z}_{t,\boldsymbol{\xi}}) \right\|_{\boldsymbol{\xi}}^2 &= O_p \left(n^{-2} \sum_{t=1}^n \sum_{s=1}^n \sqrt{\text{Tr}(\boldsymbol{\Gamma}_{\boldsymbol{\xi}}(s, s)) \text{Tr}(\boldsymbol{\Gamma}_{\boldsymbol{\xi}}(t, t))} |s - t|^{-d_{\boldsymbol{\xi}}} \right) \\ &= O_p \left(n^{-1} \max_t \text{Tr}(\boldsymbol{\Gamma}_{\boldsymbol{\xi}}(t, t)) \sum_{h=-n+1}^{n-1} |h|^{-d_{\boldsymbol{\xi}}} \right) \\ &= O_p(n^{-1}). \end{aligned} \tag{S.32}$$

where the last equality follows from the fact that $\text{Tr}(\boldsymbol{\Gamma}_{\boldsymbol{\xi}}(t, t)) \leq \mathbb{E} \|\mathbf{z}_{t,\boldsymbol{\xi}}\|_{\boldsymbol{\xi}}^2 = \mathbb{E} d_M(\mathbf{x}_t, \boldsymbol{\xi})^2 \leq (2R/(2 + \eta))^2$. Finally, since $\|(\text{dExp}_{\boldsymbol{\mu}})_{\mathbf{v}}\|_{op} = O(1)$ by Lemma S.3, (S.30) follows from (S.31) and (S.32), which completes the proof. $\square$

Remark S.4. Compared to Proposition 2, Theorem 2 derived a faster rate of convergence for the Fréchet mean not only through stronger assumptions but a more delicate analysis. The proof bounds the gradient of the difference between population and empirical loss functions $(\nabla(g_n - f_n))$ by connecting to the autocovariance functions of the data lifted at nearby points $\mathbf{z}_{t,\boldsymbol{\xi}} = \text{Log}_{\boldsymbol{\xi}} \mathbf{x}_t$.

PROOF OF PROPOSITION 3. Since $\hat{\mathcal{L}}$ and $\boldsymbol{\Lambda}$ are operators acting on finite-dimensional vector spaces, the proof is similar to Theorem 1 of Lam and Yao (2012). The major distinction is that $\hat{\mathcal{L}}$ and $\boldsymbol{\Lambda}$ are defined on different tangent spaces, so parallel transport is needed to compare these operators.

Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$ be the eigenvalues of $\boldsymbol{\Lambda}$. We have shown in the proof of Theorem 1 that $\lambda_r \geq \kappa^2$ and $\lambda_i = 0$ for $i > r$. Now consider $(\hat{\lambda}_i, \hat{\mathbf{a}}_i)$, $i = 1, 2, \dots, p$, the (eigenvalue,

eigenvector) pairs of $\hat{\mathcal{L}}$. Then $(\hat{\lambda}_i, \hat{\mathbf{b}}_i)$, $i = 1, 2, \dots, p$, are the eigenpairs of $\Phi_{\hat{\mu}, \mu} \hat{\mathcal{L}}$, where $\hat{\mathbf{b}}_i = \mathcal{P}_{\hat{\mu} \rightarrow \mu} \hat{\mathbf{a}}_i$. Since $\|\Phi_{\hat{\mu}, \mu} \hat{\mathcal{L}} - \Lambda\|_{op} = O_p(n^{-1/2})$ by (10), (S.26), and the proof of Theorem 2, the desired result follows from the proof for Theorem 1 of Lam and Yao (2012) with $\tilde{\mathbf{M}}$ (in their notation) defined as $\sum_{h=1}^{h_0} (\Phi_{\hat{\mu}, \mu} \hat{\mathcal{S}}(h)) \mathbb{E}(\mathbf{z}_{t-h} \otimes \mathbf{z}_t)^*$. $\square$

S.2 Elements of the Bures–Wasserstein manifold

In this section, we collect some facts about the Bures–Wasserstein manifold, following primarily Malagò et al. (2018). For a more comprehensive treatment, including the connection to the optimal transport between multi-dimensional Gaussian measures, we refer the readers to Malagò et al. (2018); Takatsu (2011); Ambrosio et al. (2008); Chewi et al. (2025). See also Helstrom (1967) and Bures (1969) for a motivation from quantum information theory.

The space of $q \times q$ symmetric positive-definite matrices, equipped with the Bures–Wasserstein metric, can be viewed as a p -dimensional Riemannian manifold M with $p = q(q+1)/2$. Fix an SPD matrix $\Sigma \in M$. The tangent vectors can be identified with the set of $q \times q$ symmetric matrices, and the Riemannian metric is given by

$$\langle \mathbf{U}, \mathbf{V} \rangle_{\Sigma} = \text{Tr}(\mathcal{L}_{\Sigma}(\mathbf{U}) \Sigma \mathcal{L}_{\Sigma}(\mathbf{V})) = \frac{1}{2} \text{Tr}(\mathcal{L}_{\Sigma}(\mathbf{U}) \mathbf{V}),$$

where $\mathbf{U}, \mathbf{V}$ are symmetric matrices and $\mathcal{L}_{\Sigma}(\mathbf{U})$ denotes the solution $\mathbf{X}$ to the Lyapunov equation $\mathbf{U} = \mathbf{X}\Sigma + \Sigma\mathbf{X}$. For $\Sigma_1, \Sigma_2 \in M$, the geodesic distance induced by the above metric is $d_M(\Sigma_1, \Sigma_2) = \{\text{Tr}(\Sigma_1 + \Sigma_2 - 2(\Sigma_1^{1/2} \Sigma_2 \Sigma_1^{1/2})^{1/2})\}^{1/2}$, which corresponds to the Wasserstein distance between two mean-zero q -dimensional Gaussian distributions with covariance matrices Σ_1, Σ_2 respectively.

It can be shown that if $\Sigma = \mathbf{P}\Lambda\mathbf{P}^{\top}$ is the spectral decomposition of Σ with orthonormal

$\mathbf{P}$ and $\mathbf{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_q)$, then the symmetric matrices

$$\{\mathbf{E}^{(i,j)} : i = 1, 2, \dots, q; \quad j = i, i+1, \dots, q\}$$

form an orthonormal basis for $T_{\Sigma}M$, where $\mathbf{E}^{(i,j)} = \mathbf{P}\mathbf{S}^{(i,j)}\mathbf{P}^\top$. Here, $\mathbf{S}^{(i,j)}$ is the matrix with $\sqrt{\lambda_i + \lambda_j}$ at the (i, j) -th and (j, i) -th entries and zero otherwise when $i \neq j$. When $i = j$, $\mathbf{S}^{(i,j)}$ is the matrix whose (i, i) -th element is $2\sqrt{\lambda_i}$ and zero otherwise. Indeed, for a symmetric $\mathbf{U} = (u_{ij})_{1 \leq i, j \leq q}$, the equation $\mathbf{\Lambda}\mathbf{X} + \mathbf{X}\mathbf{\Lambda} = \mathbf{U}$ admits the closed form solution $\mathbf{X} = (x_{ij})_{1 \leq i, j \leq q}$, where $x_{ij} = u_{ij}/(\lambda_i + \lambda_j)$. It follows that $\mathcal{L}_{\Sigma}(\mathbf{E}^{(i,j)}) = \mathbf{P}\tilde{\mathbf{S}}^{(i,j)}\mathbf{P}^\top$, where $\tilde{\mathbf{S}}^{(i,j)}$ is the zero matrix with the (i, j) -th and (j, i) -th entries replaced by $1/\sqrt{\lambda_i + \lambda_j}$ if $i \neq j$, and (i, i) -th entry replaced by $1/\sqrt{\lambda_i}$ if $i = j$. Thus $\langle \mathbf{E}^{(i,j)}, \mathbf{E}^{(k,l)} \rangle_{\Sigma} = \delta_{i,k}\delta_{j,l}$ where $\delta_{a,b} = 1$ if $a = b$ and zero otherwise.

The exponential map at Σ is given by

$$\text{Exp}_{\Sigma}(\mathbf{U}) = \Sigma + \mathbf{U} + \mathcal{L}_{\Sigma}(\mathbf{U})\Sigma\mathcal{L}_{\Sigma}(\mathbf{U}),$$

for symmetric matrices $\mathbf{U}$ such that $\mathbf{I}_q + \mathcal{L}_{\Sigma}(\mathbf{U})$ is SPD. Since the geodesic cannot extend indefinitely, it follows from the Hopf-Rinow theorem that M is not complete. Within the radius of injectivity, the Riemannian log map is

$$\text{Log}_{\Sigma}(\mathbf{Z}) = (\mathbf{Z}\Sigma)^{1/2} + (\Sigma\mathbf{Z})^{1/2} - 2\Sigma.$$

Parallel transport along a geodesic $t \mapsto \Sigma(t)$ in general admits no closed-form formula which can only be computed numerically by solving matrix differential equation (see Sec. 7.5 of [Malagò et al., 2018](#)). Details are omitted.

S.3 Simulation study: SPD matrix time series with the Bures–Wasserstein metric

Here we consider the manifold M as the Bures–Wasserstein manifold of $q \times q$ symmetric positive-definite (SPD) matrices equipped with the Bures–Wasserstein metric. The data are generated as in Section 4, with the following modifications. First, two reference points, $\boldsymbol{\mu}_1 = \mathbf{I}_q$ and $\boldsymbol{\mu}_2 = 5\boldsymbol{\Sigma}_q$, will be considered, where $\boldsymbol{\Sigma}_q = (0.8^{|i-j|})_{1 \leq i, j \leq q}$ has the Toeplitz structure. For the factor process (16), we set $r = 5$ and $s = 1.5$. Therefore, the signal-to-noise ratio is jointly determined by α and σ_z , for which we choose $\alpha \in \{0.2, 0.8\}$ and fix $\sigma_z = 1$. The case $\alpha = 0.2$ is referred to as the low signal case, and $\alpha = 0.8$ as the high signal case. Note that the manifold dimension is $p = q(q+1)/2$ and an orthonormal basis for tangent spaces in the Bures–Wasserstein manifold is given in Section S.2.

The pseudo-predictions of RFM are constructed similarly as in Section 4. For LFM, since each $\mathbf{x}_t$ is an SPD matrix, without reference to the nonlinear manifold, we apply LFM directly to the (half-vectorized) observed data

$$\text{vech}(\mathbf{x}_t) = (x_{t11}, x_{t21}, \dots, x_{tq1}, x_{t22}, \dots, x_{tq2}, \dots, x_{tqq})^\top.$$

Specifically, the pseudo-predictions for LFM are computed as $\text{vech}(\tilde{\mathbf{x}}_t^{(L)}) = \tilde{\boldsymbol{\mu}} + \tilde{\mathbf{A}}\tilde{\mathbf{A}}^\top \tilde{\mathbf{z}}_t$, where $\tilde{\mathbf{z}}_t = \text{vech}(\mathbf{x}_t) - \tilde{\boldsymbol{\mu}}$, $\tilde{\boldsymbol{\mu}}$ is the Euclidean mean of $\text{vech}(\mathbf{x}_1), \dots, \text{vech}(\mathbf{x}_n)$, and $\tilde{\mathbf{A}} \in \mathbb{R}^{p \times k}$ is the loading matrix estimated by LFM based on observations $\mathbf{x}_1, \dots, \mathbf{x}_n$ with k extracted factors. Similarly to the sphere case in Section 4, the predictions of LFM are not necessarily SPD, so the pseudo-predictions are constructed by $\hat{\mathbf{x}}_t^{(L)} = \mathcal{S}_\epsilon(\tilde{\mathbf{x}}_t^{(L)})$, where $\mathcal{S}_\epsilon(\cdot)$ approximately projects a symmetric matrix to the set of SPD matrices (in Frobenius sense) by replacing its nonpositive eigenvalues in the spectral decomposition with some small $\epsilon > 0$, for which we use $\epsilon = 10^{-6}$.

Figures S.1–S.2 show the relative geodesic mean squared errors of LFM and RFM with

α equal to 0.2 and 0.8, respectively, when the reference point is set to $\boldsymbol{\mu} = \boldsymbol{\mu}_1$. For reference, we also marked the rGMSEs achieved by the “oracle” predictor which uses the true loading matrix and reference point $\boldsymbol{\mu}$ in the pseudo-predictions. Across all settings, RFM consistently outperforms LFM using the same number of factors, reducing roughly 0.06–0.12 in rGMSE when 5 factors are employed, an improvement corresponding to 17.5%–60% of the oracle level of rGMSE. In addition, RFM can achieve an rGMSE near the oracle level with $r = 5$ factors. By contrast, LFM requires many more factors to attain a similar level of rGMSE. Indeed, when $q = 10$, the rGMSE of LFM using 10 factors is still higher than the oracle level.

Since the curvature of Bures–Wasserstein manifold is not constant (Takatsu, 2011), the space near $\boldsymbol{\mu}_2$ is curved differently from that near $\boldsymbol{\mu}_1$. Hence the data would exhibit different nonlinear behavior even if the latent process has the same dependence structure in the tangent space. As shown in Figures S.3–S.4, RFM outperforms LFM by an even larger margin compared to Figures S.1–S.2. These results show that the inherent nonlinearity of the manifold substantially undermined the performance of the linear factor model, while the geometry-aware RFM, which accounts for the curved manifold, yields more effective dimension reduction.

Next, we assess how well the loading subspace is recovered for RFM. Let $\hat{\mathbf{A}}_{\hat{\mathbf{E}}} = (\hat{\mathbf{a}}_{1,\hat{\mathbf{E}}}, \dots, \hat{\mathbf{a}}_{r,\hat{\mathbf{E}}})$ be the estimated loading matrix. As discussed in Sections 2 and 3, the loading spaces should be compared after parallelly transported to the tangent space at the true reference point $\boldsymbol{\mu}$. For the Bures–Wasserstein manifold, parallelly transporting $\hat{\mathbf{a}}_j = J_{\hat{\mathbf{E}}}(\hat{\mathbf{a}}_{j,\hat{\mathbf{E}}})$ from $T_{\hat{\boldsymbol{\mu}}}M$ to $T_{\boldsymbol{\mu}}M$ requires solving a matrix differential equation (Malagò et al., 2018). Let $\tilde{\mathbf{A}}_{\mathbf{E}} = (J_{\mathbf{E}}^{-1}(\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}} \hat{\mathbf{a}}_1), \dots, J_{\mathbf{E}}^{-1}(\mathcal{P}_{\hat{\boldsymbol{\mu}} \rightarrow \boldsymbol{\mu}} \hat{\mathbf{a}}_r))$ be the matrix so obtained with parallelly transported columns of $\hat{\mathbf{A}}_{\hat{\mathbf{E}}}$. Then, the discrepancy between $\tilde{\mathbf{A}}_{\mathbf{E}}$ and $\mathbf{A}_{\mathbf{E}}$ is measured by the sine-theta distance $\rho(\tilde{\mathbf{A}}_{\mathbf{E}}, \mathbf{A}_{\mathbf{E}}) = \|\sin \Theta(\text{Im}(\tilde{\mathbf{A}}_{\mathbf{E}}), \text{Im}(\mathbf{A}_{\mathbf{E}}))\|_{op}$. Note that $\rho(\tilde{\mathbf{A}}_{\mathbf{E}}, \mathbf{A}_{\mathbf{E}})$ is between 0 and 1. It is equal to 0 if and only if the column spaces of $\tilde{\mathbf{A}}_{\mathbf{E}}$ and

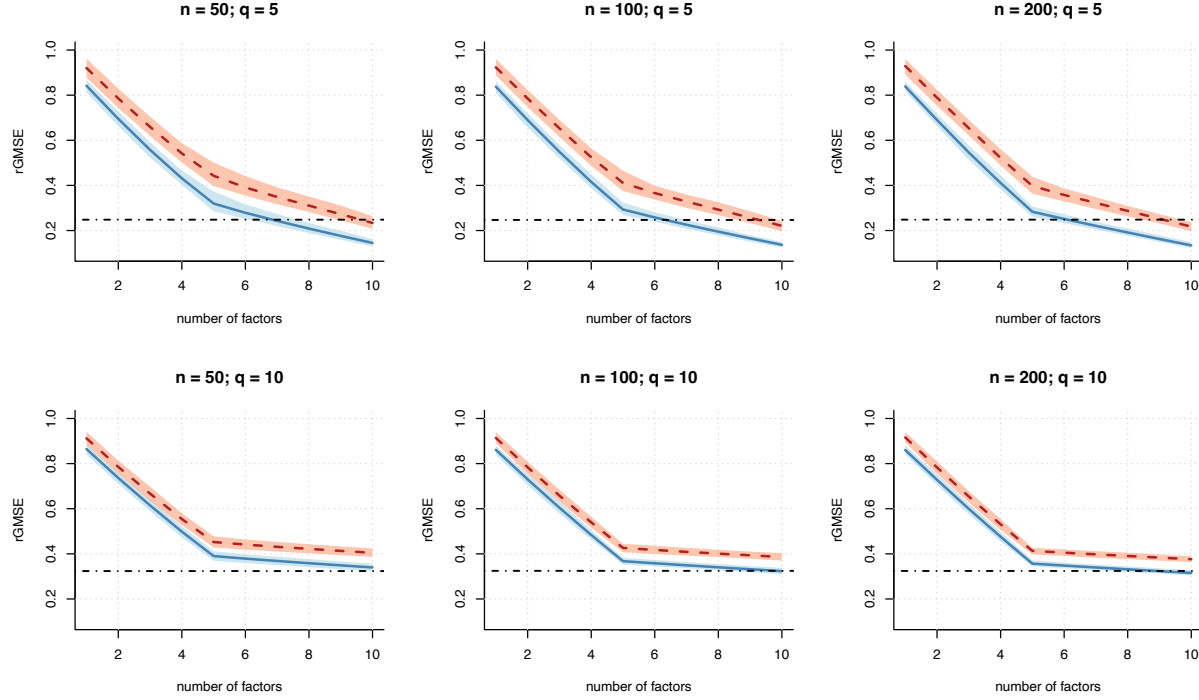


Figure S.1: Relative geodesic mean squared error (rGMSE) of the linear factor model (red dashed lines) and the Riemannian factor model (blue solid lines) for the scenario with $\boldsymbol{\mu} = \boldsymbol{\mu}_1$ and $\alpha = 0.2$. The lines are averages over 300 simulations, and the shaded areas indicate the region between the 5-th and 95-th quantiles. The horizontal dotdash line signifies the oracle level of rGMSE using the correct loading matrix and reference point.

$\mathbf{A}_{\mathbf{E}}$ are identical, and is equal to 1 if at least one principal angle is $\pi/2$.

Table S.1 reports the sine-theta distance between the estimated (after parallel transport) and the true loading spaces, showing that the discrepancy decreases with sample size. Interestingly, the convergence patterns are similar with the two reference points. Table S.2 presents the fraction of times the ratio estimator in Eq. (5) selects the correct number of factors. While accuracy is low when $n = 50$, it improves rapidly as n increases. By $n = 200$, the ratio estimator identifies the correct number of factors with high probability—nearly 100% when the matrix dimension $q = 10$, and around 70–90% when $q = 5$.

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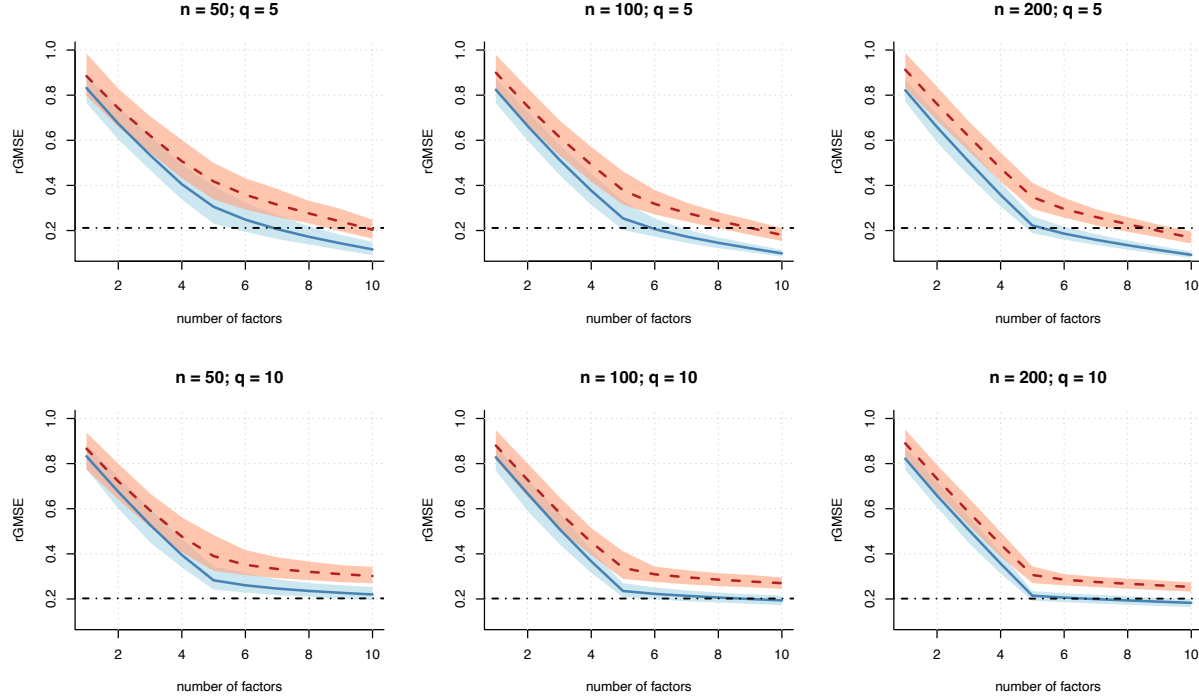


Figure S.2: Relative GMSE of the linear factor model (red dashed lines) and the Riemannian factor model (blue solid lines) for the scenario with $\boldsymbol{\mu} = \boldsymbol{\mu}_1$ and $\alpha = 0.8$. See Figure S.1 for legends.

Table S.1: Sine-theta distance between the estimated loading space (after parallel transport) and the true loading space. Standard deviations are marked in parentheses.

α	$q = 5$			$q = 10$		
	$n = 50$	$n = 100$	$n = 200$	$n = 50$	$n = 100$	$n = 200$
$\boldsymbol{\mu} = \boldsymbol{\mu}_1$						
0.2	0.59 (0.13)	0.47 (0.09)	0.40 (0.06)	0.47 (0.06)	0.37 (0.03)	0.32 (0.03)
0.8	0.75 (0.18)	0.56 (0.18)	0.37 (0.10)	0.59 (0.14)	0.39 (0.09)	0.26 (0.05)
$\boldsymbol{\mu} = \boldsymbol{\mu}_2$						
0.2	0.58 (0.11)	0.46 (0.08)	0.40 (0.07)	0.47 (0.06)	0.36 (0.03)	0.31 (0.03)
0.8	0.62 (0.18)	0.45 (0.14)	0.31 (0.08)	0.53 (0.12)	0.34 (0.06)	0.23 (0.04)

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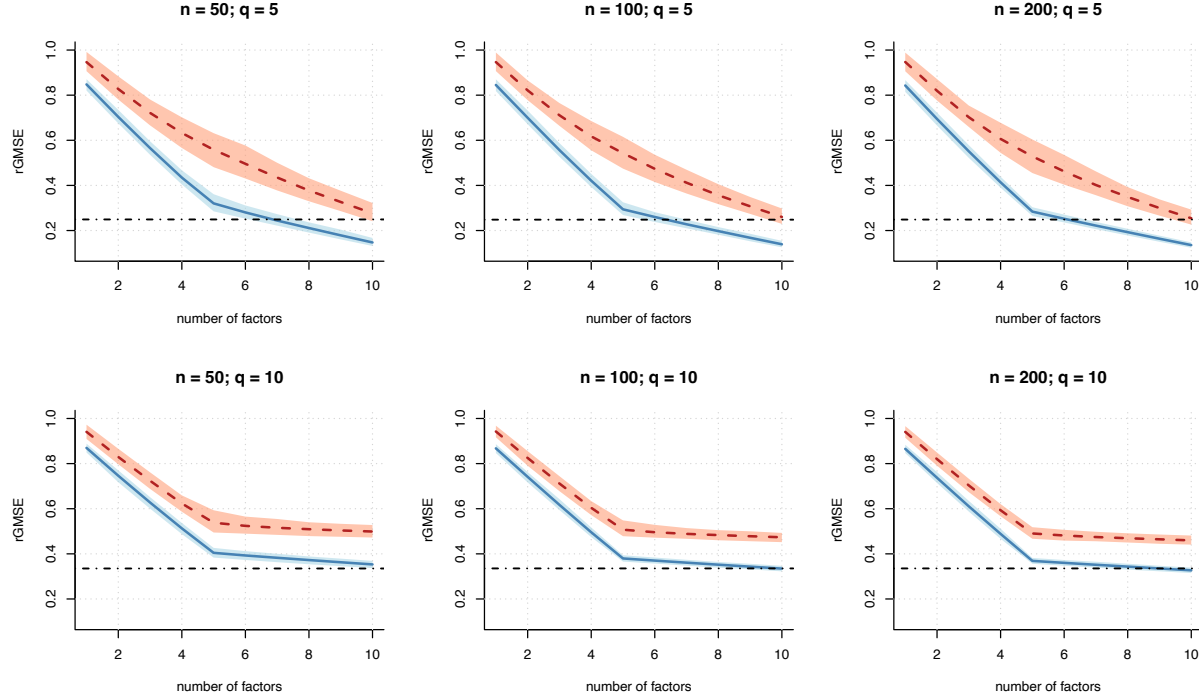


Figure S.3: Relative GMSE of the linear factor model (red dashed lines) and the Riemannian factor model (blue solid lines) for the scenario with $\boldsymbol{\mu} = \boldsymbol{\mu}_2$ and $\alpha = 0.2$. See Figure S.1 for legends.

Table S.2: Proportion of times the correct number of factors is selected by the ratio estimator (5) over the 300 simulations. Data on the Bures–Wasserstein manifold are generated according to Section S.3.

α	$q = 5$			$q = 10$		
	$n = 50$	$n = 100$	$n = 200$	$n = 50$	$n = 100$	$n = 200$
$\boldsymbol{\mu} = \boldsymbol{\mu}_1$						
0.2	0.38	0.66	0.81	0.86	0.97	1.00
0.8	0.03	0.23	0.72	0.11	0.58	0.97
$\boldsymbol{\mu} = \boldsymbol{\mu}_2$						
0.2	0.39	0.65	0.83	0.83	0.97	1.00
0.8	0.08	0.39	0.90	0.15	0.67	0.98

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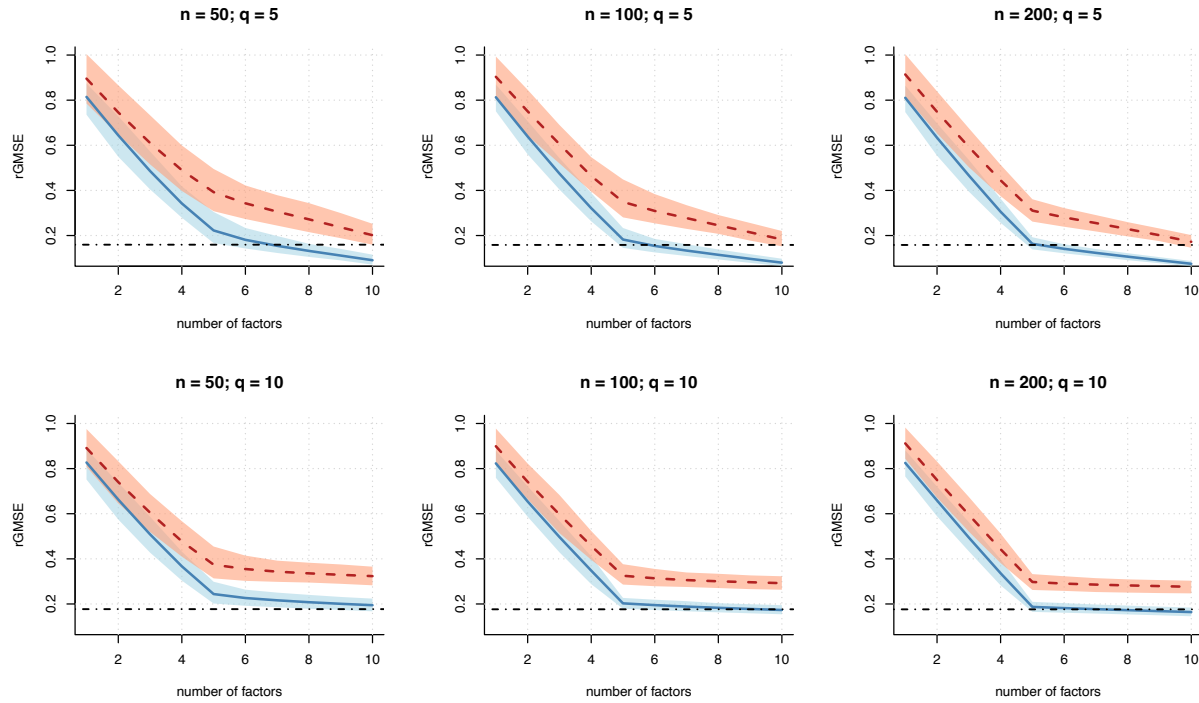


Figure S.4: Relative GMSE of the linear factor model (LFM) and the Riemannian factor model (RFM) for the scenario with $\boldsymbol{\mu} = \boldsymbol{\mu}_2$ and $\alpha = 0.8$. See Figure S.1 for legends.

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